Morarji Desai Residential School

SC-32 Bahaddurghatta (Kogunde) · Chitradurga District — 577519

Lesson Plans

Level 3 · Classes 8, 9, 10 · 72 lessons · 40-minute periods

Prepared by

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Lesson 1.01 · CONCEPT · 40 MINUTES

Functions Introduction

I CAN Analyze how a function and identify its components: name, parameters, body · Write a simple function with no parameters and call it from main program · Explain why functions help avoid code repetition and improve readability ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students, imagine you are given a task to draw a house 20 times on the board. Would you draw every single line from scratch each time, or would you create a template and reuse it? Today we learn functions, which are like reusable templates in programming. Think of a function like a recipe card. You write the recipe once, and whenever you want to make that dish, you just follow the card. You do not have to remember every step each time.	A recipe card? So instead of writing the same code again and again, I write it once and call it whenever I need? That would save so much time! Is it like a shortcut?	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let us try something together. I will write a small program without functions first. Look at this code that prints the school morning prayer three times. Now I will show you how to write the same thing using a function. You can see that the code inside the function is written only once but executed three times. Try to guess what will happen when I call the function name. Discuss with your partner and write down your prediction before I run it.	Oh I see! The print statements are inside the function but the function name is called three times. So the computer goes to the function, runs the code, comes back, and does it again. My prediction is it will print the prayer three times just like before.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A function in Python is a named block of code that performs a specific task. You define a function using the keyword def followed by the function name and parentheses. The code inside the function is indented. To execute the function you call it by writing its name followed by parentheses. The function definition tells the computer what to do and the function call tells the computer when to do it. Functions follow the DRY principle which stands for Do Not Repeat yourself. In our school we use functions to organize our programs just like how our school day is organized into periods each with a specific purpose.	So def creates the function and then we use the name with brackets to run it. The indentation shows which lines belong to the function. This is much cleaner than writing the same code everywhere. I can see how this helps when programs get longer.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Now let us apply this to something from our daily life at our school. Write a function called morning_assembly that prints three steps: stand in line, sing prayer, listen to announcements. Call this function three times to represent Monday, Tuesday, and Wednesday assemblies. Then challenge yourself: write a second function called lunch_routine that	I wrote the morning_assembly function and called it three times. Then I created lunch_routine with wash hands, collect food, eat quietly, return plate. When I call both functions the output shows assembly steps then lunch steps. The functions are like different routines in our school day.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	prints the steps of our lunch time. Call both functions and observe the output. Notice how each function has its own task and they work independently. Modify: change one parameter/block and predict outcome before running.		
Evaluate *5 min*	I will give you a short program that has repeated code. Your task is to identify the repeated part and convert it into a function. Then call the function the correct number of times. I will check if your function definition is correct, if the indentation is proper, and if your function calls produce the expected output. You have ten minutes to complete this. Who can explain why using a function here is better than the original code? Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	I converted the repeated print statements into a function called greet_student. I called it four times for four students. Using a function is better because if I need to change the greeting I only change it in one place instead of four places.	*Individual exit ticket (2 Q) at door*

Resources: Python IDLE · Textbook Chapter 1 Page 1 · our school programming worksheet handout ·

Differentiation: **App** Class 8 Foundation: Focus on understanding what a function is and writing functions with… **On** Class 9 Application: Write functions with simple print statements and call them multiple… **Adv** Class 10 Mastery: Write functions and explain the DRY principle. Compare function usage… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a function called study_time that prints three activities: read textbook, solve problems, review notes. Spiral: also recall one fact from last week.

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LESSON 1.02 · CONCEPT · 40 MINUTES

Function Parameters

I CAN Analyze the difference between parameters and arguments and distinguish between the two terms · Write functions that accept one or more parameters and pass values when calling · Explain how parameters make functions flexible and reusable with different data ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Yesterday we learned how to write functions. But notice that our functions always did exactly the same thing. Today we learn parameters which let us customize what a function does each time we call it. Think of a water tap. The tap is the function. The handle is the parameter. When you turn the handle to different positions you get different amounts of water. The tap stays the same but the result changes based on what you give it.	So parameters are like inputs to the function? If I write a function to greet a student I can pass different names as parameters and it will greet each student differently? That makes the function much more useful!	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Look at this function I wrote yesterday: def say_hello which just prints Hello Student. Now I will add a parameter: def say_hello name and inside it prints Hello plus the name. When I call say_hello with Arjun it prints Hello Arjun. When I call it with Priya it prints Hello Priya. Try changing the name inside the parentheses and predict what will print. Write three different calls with three different names from our class.	I tried say_hello with Ravi and it printed Hello Ravi. Then I tried with our school and it printed Hello our school. The function works with any text I put in the parentheses. It is like filling in a blank in a sentence.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	A parameter is a variable listed in the function definition. It is a placeholder that holds the value you pass to the function. An argument is the actual value you send when you call the function. So name is the parameter and Arjun is the argument. You can have multiple parameters separated by commas. For example def add a b takes two parameters and you call it with add 5 3. The function receives 5 in a and 3 in b. Parameters make functions flexible because the same function can work with different values. This is like how our school timetable has the same periods but different subjects each day.	So parameter is the name in the definition and argument is the value we actually pass. Multiple parameters let the function handle more complex tasks. The function definition is the blueprint and the function call with arguments is the actual building.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a function called calculate_area that takes length and width as parameters and prints the area. Test it with the dimensions of your classroom, your textbook, and your slate. Then write a function called travel_time that takes distance and speed as parameters and calculates the time taken. Use it to find how long it takes to walk from the dormitory to the school building. Discuss with your partner: what happens if you call a function with fewer arguments than parameters?	My calculate_area function works with any two numbers. I tested it with classroom dimensions 8 meters by 6 meters and got 48 square meters. My textbook is 25 centimeters by 20 centimeters and the area is 500 square centimeters. When I tried calling the function with only one argument Python gave an error saying one positional argument is missing.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a function definition with two parameters. You must write the correct function call with appropriate arguments. Then I will give you a function call and you must write the function definition that would make that call work. Pay attention to the order of arguments because they must match the order of parameters. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	I wrote the function definition correctly. I noticed that the order matters because def greet name age is different from def greet age name. The first argument goes to the first parameter and so on.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 1 Page 3 · our school parameter practice worksheet ·

Differentiation **App** Class 8 Foundation: Work with single parameter functions only. Use predefined function… **On** Class 9 Application: Write functions with two parameters and test with different values.… **Adv** Class 10 Mastery: Write functions with multiple parameters and explore default parameter… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a function called bmi_calculator that takes weight in kilograms and height in meters as parameters and prints the BMI. Spiral: also recall one fact from last week.

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LESSON 1.03 · CONCEPT · 40 MINUTES

Return Values

I CAN Use the return keyword to send a result back from a function to the calling code · Store the returned value in a variable and use it in further calculations · Compare functions that print results versus functions that return results ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students imagine you ask your friend to calculate the total cost of items in the tuck shop. There are two ways your friend can respond. First way: your friend shouts out the total to everyone. Second way: your friend quietly tells you the total and you decide what to do with it. The first way is like printing inside a function. The second way is like returning a value. Today we learn the return statement which lets a function send a result back to whoever called it.	So return is like getting an answer back from the function instead of just displaying it? That means I can use the answer in other calculations. Like if the function gives me the total cost I can then check if I have enough pocket money to pay for it.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me show you two versions of a function. First version: def add a b which prints a plus b. Second version: def add a b which returns a plus b. When I call the first version it shows the result but I cannot save it. When I call the second version with result equals add 5 3 the result variable now holds the value 8. I can use this 8 in another calculation like result times 2. Try both versions and write down the differences you observe.	The print version shows 8 on screen but result is empty. The return version does not show anything but result holds 8. I can then print result and get 8 or do result plus 10 to get 18. Return is much more useful because I can reuse the value.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	The return statement ends the function execution and sends a value back to the caller. The returned value can be stored in a variable for later use. A function can return numbers strings booleans or even lists. If a function has no return statement it automatically returns None. You should use return when the function is calculating something that other parts of your program need. You should use print when you just want to display information to the user. Most useful functions return values rather than printing them directly.	So return gives the value back to the code that called the function. Print just shows it on screen. I should use return when I need the result later and print when I just want to display something. Functions that return values are like calculators and functions that print are like announcements.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write a function called convert_temperature that takes a temperature in Celsius and returns the value in Fahrenheit using the formula F equals C times 9 divided by 5 plus 32. Store the result and print it with a message. Then write a function called calculate_grade that takes a percentage and returns the grade A B C or D based on school grading rules. Test both functions with at least three different values each. Challenge: can you write a function that returns two values at once?	My convert_temperature function takes 100 Celsius and returns 212 Fahrenheit. My calculate_grade function takes 85 and returns A, takes 65 and returns B, and takes 45 and returns C. For the challenge I learned that I can return two values by writing return min_value max_value and Python puts them in a tuple automatically.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	Modify: change one parameter/block and predict outcome before running.		
Evaluate *5 min*	I will give you a problem description. You must write a function that takes the right parameters and returns the correct result. Then I will give you some test values and you must tell me what the function will return without running the code. This tests whether you understand how return values work. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	I can predict the return value by following the formula step by step. For example if the function returns a times 2 plus 1 and I call it with 5 the answer is 11. Tracing through the function line by line helps me get the right answer.	*Individual exit ticket (2 Q) at door*

Resources: Python IDLE · Textbook Chapter 1 Page 5 · our school return value practice sheet ·

Differentiation: **App** Class 8 Foundation: Focus on functions that return a single simple value. Use tracing… **On** Class 9 Application: Write functions that return values and use those values in… **Adv** Class 10 Mastery: Write functions returning multiple values using tuples. Explore how… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a function called calculate_average that takes three test scores as parameters and returns the average. Spiral: also recall one fact from last week.

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LESSON 1.04 · CONCEPT · 40 MINUTES

Arrays Basics

I CAN Create and manipulate a list in Python and understand that lists can hold multiple values in one variable · Access individual elements using index numbers starting from zero · Modify list elements by assigning new values to specific index positions ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students imagine you have ten pencils in your pencil box. You can hold all ten in one box instead of carrying ten separate boxes. Today we learn about lists in Python which work like that pencil box. A list lets you store many values under one name. Instead of creating ten separate variables for ten students marks you put all ten marks in one list. This makes your program much easier to manage.	So a list is like a container that holds many items? I can store all my subject marks in one list instead of creating separate variables for each subject. That saves a lot of time and makes the code shorter.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me create a list of five school houses: houses equals opening square bracket Arya Bharati Chandrika Drona Ekalavya closing square bracket. Now I will print the first house using houses square bracket zero. Notice we start counting from zero not one. Try to predict what houses square bracket 2 will give you. Then try to access houses square bracket 5 and see what happens. Discuss with your partner why Python starts counting from zero.	houses square bracket 2 gives Chandrika because we count zero one two. houses square bracket 5 gives an error because we only have five houses numbered zero to four. Starting from zero must be because computers count from zero. It is like floors in some buildings where the ground floor is zero.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	A list in Python is created using square brackets with values separated by commas. Each element has an index starting from zero. The first element is at index zero the second at index one and so on. You access an element using the list name followed by the index in square brackets. You can change an element by assigning a new value to its index position. Lists can contain different data types including numbers strings and even other lists. The length of a list tells you how many elements it contains. Indexing from zero is universal in programming and you must always remember this.	So square brackets create the list and square brackets with an index access an element. Zero based indexing means the first element is always at position zero. I can also change any element by assigning a new value to its position. Lists are very flexible containers for storing groups of related data.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a list called attendance that holds the attendance status present or absent for all students in your group for today. Then create a list called subjects that holds the names of all your subjects today. Access and print the third subject. Change the second attendance record from absent to present. Now create a list called marks that holds five test scores and find which score is the highest by comparing elements.	My attendance list has present for six students and absent for one. My subjects list has Mathematics Science English Kannada and Social Studies. The third subject at index two is English. I changed Ravi attendance from absent to present. For finding the highest mark I compared each element and found that 92 at index three is the highest.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a list and ask you questions about specific elements using index numbers. You must also write code to create a list from a description and access certain elements. I will check if you use correct zero-based indexing and if you can modify elements correctly. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	I used correct indexing. I noticed that practice is needed to remember that the third element is at index two not index three. Writing the index number above each element helps me keep track.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 1 Page 7 · our school list practice worksheet ·

Differentiation **App** Class 8 Foundation: Focus on creating simple lists and accessing elements by index. Use… **On** Class 9 Application: Create lists and perform basic operations including access… **Adv** Class 10 Mastery: Create nested lists and explore advanced indexing including negative… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a list called weekly_menu that holds the lunch menu for Monday through Friday. Spiral: also recall one fact from last week.

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Lesson 1.05 · concept · 40 minutes

Array Operations

I can Use append and insert to add elements to a list at specified positions · Use remove and pop to delete elements from a list · Use the len function and the in operator to check list properties ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Yesterday we created lists and accessed elements. But what happens when you need to add a new student to your attendance list or remove a student who has left? Today we learn list operations which let us add and remove elements. Think of your list as a train with compartments. You can add new compartments at the end or insert one in the middle or remove one from anywhere.	So lists are not fixed. I can change their size by adding or removing elements. That makes them much more useful than regular variables. I can build a list step by step as I collect data.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	I have a list called fruits with banana orange and apple. Let me add mango at the end using fruits dot append mango. Now the list has four items. Let me insert grape at position one using fruits dot insert 1 grape. Now grape is between banana and orange. Let me remove orange using fruits dot remove orange. Try each operation and print the list after each change to see what happens.	After append the mango appears at the end. After insert the grape pushes the other items to make room. After remove the orange is gone and the remaining items shift to fill the gap. The list automatically adjusts its size.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The append method adds an element to the end of the list. The insert method adds an element at a specific index position. The remove method deletes the first occurrence of a specified value. The pop method removes and returns the element at a given index or the last element if no index is given. The len function returns the number of elements in the list. The in operator checks if a value exists in the list and returns True or False. These operations make lists dynamic meaning they can grow and shrink as needed during program execution.	So append adds to the end insert adds at a position remove deletes by value and pop deletes by position. Len tells me how many items are in the list and in checks if something is there. These operations let me manage the list like a living collection that changes as needed.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a program that manages a residential class roster. Start with an empty list. Add five student names using append. Insert a new student at position two. Remove a student who transferred to another school. Print the length of the list after each operation. Check if a specific student is in the list using the in operator. Finally print the complete updated roster. Challenge: use pop to remove the last student and store their name in a variable.	My roster started empty. I added Arjun Priya Ravi Kavya and Meena. Then I inserted Sanjay at position two. After that I removed Kavya because she transferred. The list length changed from 5 to 6 to 5. Sanjay is in the list returns True. I popped Meena and stored her name. The final roster has five students.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a sequence of operations to perform on a list. You must write the code for each operation and predict the state of the list after each step. Then I will give you a final list state and ask what operations could have produced it.	I can track the list state after each operation by writing it down. The key insight is that insert shifts elements to the right and remove shifts elements to the left to fill gaps.	*Individual exit ticket (2 Q) at door*

Resources: Python IDLE · Textbook Chapter 1 Page 9 · our school list operations cheat sheet ·

Differentiation: **App** Class 8 Foundation: Practice append and remove operations with teacher guided… **On** Class 9 Application: Perform all basic list operations independently. Create a student… **Adv** Class 10 Mastery: Combine multiple list operations to solve complex problems. Explore… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that creates a to-do list for your evening study. Spiral: also recall one fact from last week.

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LESSON 1.06 · CONCEPT · 40 MINUTES

String Operations

I CAN Perform string concatenation and repetition to combine and repeat text · Use string methods like upper lower strip and replace to transform text · Slice strings to extract substrings using index ranges ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students you use strings every day when you write your name read signs and send messages. Today we learn how to manipulate strings in Python. Imagine you have a rubber band. You can stretch it join two rubber bands together or cut a piece from it. Strings work similarly in programming. You can join two strings stretch a string by repeating it or cut out a piece using slicing.	So strings are not just text. They are like objects I can work with. I can combine them change their case and even cut pieces out. That means I can process text data in my programs just like I process numbers.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me join two strings: first_name equals our school and last_name equals Academy. Full_name equals first_name plus space plus last_name. Now I have Greenwood Academy. Let me repeat a string: dash equals star times 20 gives me twenty stars. Let me change case: hello dot upper gives HELLO and HELLO dot lower gives hello. Try these operations and create your own joined and repeated strings.	I joined my first name and last name with a space between them. I repeated the character equals sign twenty times to make a line. upper changed everything to capital letters and lower changed everything to small letters. These methods are very useful for formatting text.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	String concatenation uses the plus operator to join two strings. String repetition uses the multiply operator to repeat a string multiple times. The upper method converts all characters to uppercase. The lower method converts to lowercase. The strip method removes leading and trailing spaces. The replace method substitutes one substring with another. String slicing uses the syntax string start colon end to extract a portion. The start index is included but the end index is excluded. Slicing does not modify the original string but returns a new one.	Plus joins strings multiply repeats them and dot methods transform them. upper and lower change case strip removes spaces and replace swaps text. Slicing with colon extracts a piece. Strings are immutable which means the original string never changes. Every operation creates a new string.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a program that formats a residential student ID card. Use string concatenation to combine the student name school name and class into a formatted string. Use upper to make the school name stand out. Use strip to clean up any extra spaces in student names. Use replace to change a placeholder name to the actual student name. Use slicing to extract the first three letters of the student name as initials. Print the complete formatted ID card.	My ID card shows our school ACADEMY in uppercase at the top. The student name is Priya Sharma and her class is Class 9 Section A. I used strip to remove extra spaces from the name input. I used replace to swap placeholder with Priya Sharma. The initials are the first three letters P from Priya. The formatted output looks professional and clean.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a string and ask you to perform specific operations on it. You must write the correct method calls and predict the result. I will also give you a desired output and you must figure out what string operations produced it.	I can trace through string operations step by step. For example if the string is hello world then strip removes spaces dot upper makes it HELLO WORLD and replace changes O to zero giving HELLO W0RLD. Practicing these operations builds confidence.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 1 Page 11 · our school string methods reference card ·

Differentiation **App** Class 8 Foundation: Focus on concatenation upper lower and strip methods. Use predefined… **On** Class 9 Application: Combine multiple string operations to format text output. Work on… **Adv** Class 10 Mastery: Use string methods in combination with list operations for complex… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that takes your full name and formats it in three ways: FULL NAME in all capitals, full name in all lowercase, and First M. Spiral: also recall one fact from last week.

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Lesson 1.07 · concept · 40 minutes

Nested Loops

I can Write a loop inside another loop to process multi-dimensional data patterns · Predict the output of nested loops by tracing through iterations step by step · Apply nested loops to solve real problems like multiplication tables and grid patterns ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students in our multiplication table we have rows and columns. Each row is a multiple and each column is a multiplier. To fill the entire table you need a loop inside a loop. Today we learn nested loops which are loops within loops. Think of it like this: the outer loop picks which row you are on and the inner loop goes through all the columns in that row before moving to the next row.	So the outer loop runs once for each row and the inner loop runs completely for each iteration of the outer loop. That means if the outer loop runs 5 times and the inner loop runs 3 times the total iterations are 15. The inner loop finishes everything before the outer loop moves forward.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me write a nested loop. The outer loop uses i from 1 to 3 and the inner loop uses j from 1 to 3. Inside the inner loop I print i times j. Watch what happens: when i equals 1 the inner loop runs three times printing 1 times 1 equals 1 then 1 times 2 equals 2 then 1 times 3 equals 3. Then i becomes 2 and the inner loop runs again. Draw a table on paper showing which values of i and j produce which outputs.	I drew the table. When i is 1 the outputs are 1 2 3. When i is 2 the outputs are 2 4 6. When i is 3 the outputs are 3 6 9. The inner loop runs three times for each value of the outer loop. Total print statements are 9 which is 3 times 3.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A nested loop is a loop inside another loop. The outer loop controls the number of times the inner loop repeats entirely. For each single iteration of the outer loop the inner loop runs through all its iterations. The total number of iterations is the outer count multiplied by the inner count. Nested loops are essential for processing 2D data like grids tables and matrices. They are also used for pattern printing and coordinate systems. Be careful with nested loops because they can become slow if both loops have large ranges.	So the outer loop is like the teacher assigning rows and the inner loop is like students filling each row. The inner loop must finish completely before the outer loop moves on. The total work is outer count times inner count. I need to be careful with large ranges because nested loops do a lot of work.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a program that prints a multiplication table from 1 times 1 to 5 times 5 using nested loops. Format the output so numbers align in columns. Then write a program that prints a right triangle of stars where row 1 has 1 star row 2 has 2 stars and so on up to 5 rows. Challenge: print an inverted triangle where row 1 has 5 stars and row 5 has 1 star.	My multiplication table prints neatly in columns. The triangle program uses the outer loop for rows and the inner loop prints j stars for row j. For the inverted triangle I print 6 minus j stars for each row. The nested loop pattern is the same but the calculation inside changes the shape.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a nested loop program and ask you to trace through it completely. You must fill in a table showing the values of i and j at each step and the output produced. Then I will give you a pattern and you must write the nested loop code that produces it. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	Tracing nested loops requires patience. I write down each value of i then write all values of j for that i before moving to the next i. This systematic approach helps me avoid mistakes.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Trace simple nested loops with small ranges like 1 to 3. Use… **On** Class 9 Application: Write nested loops independently to create multiplication tables… **Adv** Class 10 Mastery: Create complex grid patterns and multi-dimensional data processing… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a nested loop program that prints the table of 7 from 7 times 1 equals 7 to 7 times 10 equals 70. Spiral: also recall one fact from last week.

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LESSON 1.08 · CONCEPT · 40 MINUTES

Exception Handling

I CAN Analyze and compare common Python exceptions including TypeError ValueError and ZeroDivisionError · Use try except blocks to handle errors gracefully without crashing the program · Write programs that provide meaningful error messages to guide the user ·

Phase	Teacher Activity	Student Activity	Assessment Check
ENGAGE *5 min*	Students what happens when you try to divide a number by zero in mathematics. The teacher says it is undefined. In Python the program crashes with an error. But crashing is not good. Today we learn exception handling which lets us catch errors and handle them gracefully. Think of it like this: if you trip while walking you can either fall down or catch yourself on a railing. Exception handling is the railing that catches your program before it crashes.	So instead of the program stopping with an ugly error message I can catch the error and show a friendly message or try a different approach. That makes my programs more robust and user-friendly.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me try dividing 10 by 0 in Python. Watch what happens. The program gives a ZeroDivisionError. Now let me wrap it in a try block: try result equals 10 divided by 0 except ZeroDivisionError print cannot divide by zero. Now the program does not crash. It prints a friendly message instead. Try other errors like converting hello to a number using int and see what exception occurs.	The ZeroDivisionError was caught and my message printed. When I tried int of hello I got a ValueError. I can catch that too with except ValueError. The try block is like a safety net that catches problems and lets the program continue.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	An exception is an error that occurs during program execution. Common exceptions include ZeroDivisionError for dividing by zero TypeError for wrong data types and ValueError for invalid values. The try except block lets you attempt risky code in the try section and handle errors in the except section. If no error occurs the except section is skipped. You can have multiple except blocks for different error types. The else block runs only if no exception occurred. The finally block always runs whether an exception occurred or not. Exception handling makes programs robust and user-friendly.	Try contains code that might fail. Except catches specific errors. Else runs when there is no error. Finally runs always. Multiple except blocks handle different error types. This structure lets me write programs that do not crash unexpectedly and provide helpful feedback to users.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write a calculator program that takes two numbers from the user and performs division. Handle ZeroDivisionError when the second number is zero. Handle ValueError when the user enters text instead of numbers. Provide clear messages for each type of error. Add an else block that prints the result only when the division succeeds. Add a finally block that prints Thank you for using the calculator. Modify: change one parameter/block and predict outcome before running.	My calculator catches division by zero and shows Please enter a non-zero divisor. It catches invalid input and shows Please enter valid numbers. When the calculation works the else block prints the result. The finally block always prints the thank you message. The program never crashes no matter what the user enters.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you programs with potential errors. You must identify what type of exception might occur and write the correct try except block to handle it. You must also write meaningful error messages that help the user understand what went wrong and how to fix it. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	I can identify the error type by thinking about what could go wrong. Division by zero is ZeroDivisionError. Converting text to number is ValueError. Adding a string and number is TypeError. Each requires a specific except block.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 1 Page 15 · our school exception handling reference ·

Differentiation **App** Class 8 Foundation: Focus on try except blocks for ZeroDivisionError and ValueError... **On** Class 9 Application: Handle multiple exception types in one program. Write meaningful... **Adv** Class 10 Mastery: Implement comprehensive exception handling in complex programs.... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that asks for the student age and converts it to an integer. Spiral: also recall one fact from last week.

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Lesson 1.09 · concept · 40 minutes

File Input and Output

I can Open read and close text files using the open function and with statement · Write data to a file using write and writelines methods · Use file handling to save and retrieve program data persistently ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students when you write a program and run it the variables exist only while the program runs. When you close the program everything is lost. But what if you want to save your data like marks attendance or notes? Today we learn file handling which lets your programs save data to files and read data from files. Think of it like writing in a notebook. Your thoughts stay in the notebook even after you stop writing.	So file handling lets my program save information permanently. I can write student marks to a file and read them back later even after restarting the computer. This is how real applications store data.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me create a file and write something to it. I will use with open test dot txt in write mode as file then file dot write hello our school. When I run this a file called test dot txt is created with the text inside. Now let me read it back: with open test dot txt in read mode as file then print file dot read. Watch the text appear on screen. Try creating your own file with your name and reading it.	I created a file called my_name dot txt and wrote my name inside. When I read it back my name appeared on screen. The with statement automatically closed the file when I was done. The file still exists on the computer even after the program ended.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	File handling in Python uses the open function with a filename and mode. The write mode w creates a new file or overwrites an existing one. The read mode r opens an existing file for reading. The append mode a adds data to the end of an existing file. The with statement automatically closes the file when the block ends which is the recommended approach. The write method adds text to the file. The read method returns the entire file content as a string. The readline method reads one line at a time. The writelines method writes a list of strings to the file. Always close files properly to avoid data loss.	Open creates or accesses a file. Write mode overwrites read mode accesses and append mode adds to the end. The with statement is the safest way because it auto-closes. Write adds text read gets text and readline gets one line. Proper file handling prevents data loss and corruption.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Write a program that saves a student report card to a file. Create variables for student name class and marks in three subjects. Write all this data to a file called report dot txt in a readable format. Then write a program that reads this file and calculates the total and average marks. Save the results to a new file called results dot txt. Challenge: use append mode to add a new student record to an existing file.	My report card file has the student name class and marks formatted nicely. My results program reads the marks calculates total as 235 and average as 78.33 and saves it. For the challenge I appended a second student record without deleting the first one. The append mode preserved the existing data.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a scenario where a program needs to save and load data. You must write the complete file handling code including opening writing reading and closing files. I will check if you use the with statement correctly and handle possible errors like file not found. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	I used try except with FileNotFoundError in case the file does not exist yet. The with statement ensures the file closes properly. Writing data in a consistent format makes it easier to read back later.	*Individual exit ticket (2 Q) at door*

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Differentiation: **App** Class 8 Foundation: Focus on basic write and read operations using the with statement.... **On** Class 9 Application: Write and read files independently. Create report card programs... **Adv** Class 10 Mastery: Implement complete file-based data management systems. Use append mode... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that saves your daily study log to a file. Spiral: also recall one fact from last week.

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LESSON 1.10 · CONCEPT · 40 MINUTES

Modular Programming

I CAN Break a large program into smaller functions each handling a specific task · Understand how modules help organize code and make it reusable · Import and use built-in modules like math random and datetime ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students imagine you are building a house. You would not try to build everything yourself. You would have separate teams for plumbing electrical work and painting. Each team is an expert in their area. Modular programming works the same way. Instead of writing one long program you break it into smaller modules or functions each responsible for one task. This makes your code organized testable and reusable.	So modular programming is like dividing work among specialists. Each function becomes an expert in one task. I can reuse the same function in different programs and if there is a bug I only need to fix it in one place.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me show you two versions of a program. First version is one long main function with everything mixed together. Second version has separate functions: get_input process_data and display_output. The second version is much easier to read and understand. Now let me show you the math module: import math then print math dot sqrt 16 gives 4.0. Try importing the random module and using random dot randint to get a number between 1 and 10.	The modular version is much cleaner. Each function has a clear name that describes what it does. The math module gave me square root and the random module gave me random numbers. I can use these built-in modules without writing the math myself.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Modular programming means organizing code into separate self-contained units. Each unit handles one specific task. This follows the Single Responsibility Principle. Python modules are files containing functions and variables that can be imported into other programs. The import statement loads a module. You can import specific functions with from module import function. Built-in modules include math for mathematical functions random for random numbers and datetime for date and time operations. Modular code is easier to debug test and maintain. It also enables teamwork as different people can work on different modules.	Modules are like toolboxes. Each toolbox contains tools for a specific job. I import the toolbox and use its tools. The math module has mathematical tools. The random module has random number tools. Breaking programs into functions makes each part testable and reusable.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a residential student management program using modular design. Write separate functions for add_student view_students search_student and save_to_file. Then create a main menu that lets the user choose which function to run. Import the datetime module to timestamp each student record. Import the random module to generate	My program has four functions each handling one task. The main menu loops and lets the user choose an option. add_student collects data view_students displays all records search_student finds a specific student and save_to_file writes	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	unique student IDs. Each function should be short and focused on one task.	everything to disk. The datetime module adds timestamps and random generates IDs like STU4521.	
Evaluate *5 min*	I will give you a monolithic program with everything in one function. You must identify the logical sections and break them into separate functions. Then write the main program that calls these functions in the correct order. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	I identified four main tasks in the program and created a function for each. The main program becomes a simple sequence of function calls. Each function is short and focused making the code much easier to understand and modify.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Focus on understanding what a module is and using built-in modules... **On** Class 9 Application: Write modular programs with multiple functions and import built-in... **Adv** Class 10 Mastery: Design complex modular systems with clear separation of concerns.... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that calculates the area of a circle using the math module for the value of pi. Spiral: also recall one fact from last week.

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LESSON 1.11 · CONCEPT · 40 MINUTES

Introduction to Recursion

I CAN Analyze recursion as a function that calls itself with a modified parameter and justify your choice · Write a recursive function with a clear base case to stop the recursion · Trace through a recursive function call to understand the call stack ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students imagine you are standing between two mirrors facing each other. You see your reflection repeated infinitely. Recursion in programming is similar but with a twist. A recursive function calls itself but each time it gets closer to a stopping point. Think of it like Russian nesting dolls. Each doll contains a smaller version of itself until you reach the smallest solid doll. That smallest doll is the base case that stops the recursion.	So a recursive function calls itself but it must have a stopping condition. Without the stopping condition it would call itself forever like the mirrors. The base case is like the smallest doll that does not open. I can see how this could simplify some problems.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me write a simple recursive function. def countdown n: if n equals 0 print liftoff else print n then countdown n minus 1. When I call countdown 5 it prints 5 4 3 2 1 liftoff. Trace through this on paper. Write down each call and what happens. What would happen if I forgot the if statement. What would happen if I did not change n in each call.	I traced it. countdown 5 prints 5 then calls countdown 4. countdown 4 prints 4 then calls countdown 3. This continues until countdown 0 prints liftoff and stops. Without the if statement it would call itself forever causing a stack overflow. Without changing n the base case would never be reached.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Recursion is when a function calls itself. Every recursive function needs two parts. The base case is the condition that stops the recursion. Without it the function would call itself infinitely. The recursive case is where the function calls itself with a modified parameter that moves toward the base case. The call stack keeps track of all active function calls. Each recursive call adds a new frame to the stack. When the base case is reached the frames start returning values back up the stack. Recursion is useful for problems that have a self-similar structure like counting tree branches or calculating factorials.	Base case stops recursion recursive case moves toward base case and the call stack manages everything. Each call waits for the next call to finish. The base case is crucial. Without it Python gives a maximum recursion depth error. Recursion simplifies certain problems by breaking them into smaller versions of the same problem.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write a recursive function to calculate factorial of a number. Remember that n factorial equals n times n minus 1 factorial and 0 factorial equals 1. Test with factorial of 5 factorial of 0 and factorial of 1. Then write a recursive function to calculate the sum of numbers from 1 to n. Challenge: write a recursive function to reverse a string by taking the last character plus the reverse of the remaining characters. Modify: change one parameter/block and predict outcome before running.	My factorial function returns 120 for factorial of 5 because 5 times 4 times 3 times 2 times 1 equals 120. factorial of 0 returns 1 which is the base case. My sum function returns 15 for sum of 1 to 5. For reversing hello it takes o plus reverse of hell which continues until the string is empty returning olleh.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a recursive function and ask you to trace through it completely. You must show each function call on the stack and the return values as they propagate back. You must also identify the base case and the recursive case. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	Tracing recursion requires patience. I draw the call stack showing each level. The base case returns first then each level above uses that return value. Understanding the stack helps me predict recursive function output.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Focus on understanding the concept of a function calling itself.... **On** Class 9 Application: Write recursive functions for factorial and sum. Trace through... **Adv** Class 10 Mastery: Implement recursive solutions for complex problems including string... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a recursive function called sum_digits that takes a number and returns the sum of its digits. Spiral: also recall one fact from last week.

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LESSON 1.12 · CONCEPT · 40 MINUTES

Programming Project

I CAN Apply functions parameters return values lists and file handling in a complete project · Plan a program by identifying functions needed and how they connect together · Debug and test a multi-function program to ensure all parts work correctly ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students over the past weeks you have learned many programming tools: functions parameters return values lists loops file handling and more. Today you will combine all these skills into a complete project. This is like building a house where you use bricks cement wood and paint together. Each material alone is simple but combined they create something amazing. Your project will demonstrate how all these concepts work together.	I am excited to use everything I learned in one big project. I need to plan which functions I need and how they will work together. This feels like being a real programmer who builds complete applications.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let us plan a residential Library Management System together. First we identify the main tasks: add a book issue a book return a book display all books and search for a book. Each task becomes a function. We use a list to store book records. We use file handling to save the data. We use loops for the menu. We use string operations for formatting. Let me sketch the program structure on the board and you suggest what each function should do.	The add_book function should collect title author and availability. The issue_book function should mark a book as issued. The return_book function should mark it available again. The display function should loop through all books. The search function should find a book by title. We need a main menu loop to let the user choose which operation to perform.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A programming project requires planning before coding. Start by identifying all the functions your program needs. Write pseudocode for each function describing what it does step by step. Decide what data structures you need such as lists or dictionaries. Plan how functions will communicate through parameters and return values. Write one function at a time and test it before moving to the next. Use comments to explain complex logic. Debug by checking each function individually then testing the complete program. Documentation helps others understand your code and helps you maintain it later.	Planning saves time because I know what to build before I start coding. Testing each function separately catches errors early. Comments explain my thinking. The project structure shows how simple functions combine to create a useful application.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Build the complete our school Library Management System. Include at least five functions each handling one operation. Use a list of dictionaries to store book records with title author and availability status. Implement a menu loop that runs until the user chooses to exit. Save all book data to a file when the program ends and load it when the program starts. Add error handling for invalid menu choices and	My library system has six functions: add_book issue_book return_book display_books search_book and load_data. I use a list of dictionaries where each dictionary represents a book. The menu loops until the user chooses exit. I save data to library_data dot txt using json format. Error handling catches	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	missing books. Format the output neatly using string methods. Modify: change one parameter/block and predict outcome before running.	invalid input and missing books. The output is neatly formatted in columns.	
Evaluate *5 min*	I will evaluate your project based on these criteria: correct use of functions with parameters and return values proper use of lists and file handling clear program structure with a working menu error handling that prevents crashes and neat formatted output. Present your project to the class explaining your design choices and how each function works. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict).	My project meets all criteria. I used six separate functions with proper parameters. Lists store the data and file handling saves it. The menu loop works correctly. Error handling catches all invalid inputs. My presentation explained how I planned each function before coding.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Complete a simplified project with three functions: add display and... **On** Class 9 Application: Build the complete project with five functions and file handling.... **Adv** Class 10 Mastery: Extend the project with additional features like sorting books by... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a simple Student Grade Tracker project. Spiral: also recall one fact from last week.

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Lesson 2.01 · lab · 40 minutes

Lists Advanced

I can Use list slicing to extract sublists using start stop and step parameters · Apply list comprehension to create lists from expressions concisely · Understand list copying and the difference between shallow and reference copies ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students yesterday we learned basic list operations like append and remove. Today we go deeper into lists. Imagine you have a row of 30 students in assembly. You want to take a photograph of just students 5 through 15. That is like slicing a list. You take a portion of the list without changing the original. Advanced lists also let you create new lists from existing ones using powerful expressions.	Slicing a list means taking a portion of it. I can get students 5 through 15 without removing anyone from the original row. List comprehension sounds like a shorter way to create lists. I am curious about how that works.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a list of numbers 0 through 9. Now slice it: numbers 2 colon 7 gives me elements at indices 2 through 6. numbers colon colon 2 gives every other element. numbers colon colon minus 1 reverses the list. Now list comprehension: square_list equals bracket x times x for x in range 10 bracket. This creates 0 1 4 9 16 through 81 in one line. Try different slices and comprehension expressions.	The slice with step 2 skips every other element. Reversing with minus 1 step is clever. The list comprehension created squares of 0 through 9 in one line instead of a for loop. I can add conditions too like even numbers only. This is much faster to write.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	List slicing uses the syntax list start colon stop colon step. Start is where slicing begins inclusive. Stop is where it ends exclusive. Step determines the increment between elements. Omitting start begins at the beginning. Omitting stop goes to the end. Step of 2 takes every other element. Negative step reverses the direction. List comprehension creates a new list from an expression in one line using the pattern bracket expression for item in iterable bracket. You can add if conditions to filter elements. List copying is important because assigning one list to another creates a reference not a copy. Use copy method or slicing to create independent copies.	Slicing extracts portions with start stop and step. Comprehension builds lists concisely with expression for item in iterable. Copying requires care because simple assignment creates a reference. The copy method and slicing create actual copies that can be modified independently.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a list of marks for 20 students. Use slicing to extract marks of the first 5 students the last 5 students and every third student. Use list comprehension to create a list of pass marks where marks above 50 are marked as pass and below as fail. Use comprehension with a condition to create a list of only the failing marks. Copy the original list and	My slices give first 5 marks from index 0 to 4 and last 5 from index 15 to 19. Every third student uses step 3. My pass_fail comprehension creates PASS or FAIL for each mark. My failing marks list uses if mark less than 50. The copy was	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	modify the copy to show what happens with reference versus actual copies.	modified independently proving it is a real copy not a reference.	
Evaluate *5 min*	I will give you a list and ask you to write slicing expressions to extract specific portions. Then I will give you a description and ask you to write the list comprehension that produces it. You must also explain the difference between reference and copy assignment.	Slicing notation becomes natural with practice. The key is remembering that stop is exclusive. List comprehension is a powerful tool that replaces multi-line loops with a single expressive line.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 1 · our school advanced list exercises ·

Differentiation **App** Class 8 Foundation: Focus on basic slicing with start and stop only. Use visual index… **On** Class 9 Application: Use slicing with step parameters and negative indices. Write list… **Adv** Class 10 Mastery: Combine slicing and comprehension to solve complex problems. Explore… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a list of numbers from 1 to 30. Spiral: also recall one fact from last week.

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Lesson 2.02 · lab · 40 minutes

List Methods

I can Use sort and reverse to order list elements in ascending and descending order · Apply count and index methods to find element frequency and position · Use extend and clear methods to merge lists and empty them completely ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students in our residential school we line up in order of height for assembly. The teacher arranges us from shortest to tallest. Today we learn list methods that sort and organize data just like the teacher arranges the line. Python has built-in methods that can sort reverse count and find elements in a list.	So Python can sort lists automatically. I do not have to write sorting code from scratch. The sort method will arrange my marks from lowest to highest and I can reverse it to go from highest to lowest.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a list of 8 student marks: 78 92 45 88 63 95 72 81. Now I call marks dot sort and print the list. The marks are now in order from 45 to 95. Now I call marks dot reverse and they go from 95 to 45. Let me count how many students scored above 80 using a comprehension then use marks dot count to find how many got exactly 72. Try these methods and observe the changes.	sort rearranged the list in ascending order. reverse flipped it to descending. count told me how many times 72 appears in the list. index found the position of a specific mark. These methods modify the original list directly.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The sort method arranges elements in ascending order by default. Pass reverse equals true to sort in descending order. sort modifies the original list in place and returns None. The reverse method flips the list order. The count method returns how many times a value appears. The index method returns the position of the first occurrence of a value. The extend method adds all elements from another list to the end. The clear method removes all elements leaving an empty list. These methods are essential for data manipulation and organization.	Sort orders the list and reverse flips it. count counts occurrences and index finds position. extend merges lists and clear empties them. All these methods except clear modify the list in place. They are like tools for organizing and managing data efficiently.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a program that manages our school exam results. Store marks for 10 students in a list. Sort them to find the highest and lowest scores. Count how many students scored above 80. Find the position of the median score. Extend the list with 5 new marks from another test. Reverse the sorted list to show results from highest to lowest. Finally clear the list and print its length to confirm it is empty.	My sorted list shows lowest 35 and highest 98. Eight students scored above 80. The median is at index 4 in the sorted list. I extended with 5 more marks from the science test. Reversed shows 98 at the beginning. Clear made the list empty and length is 0. The program handles all data management tasks.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a list of data and a series of operations to perform. You must write the correct method calls and predict the output. Then I will give you a desired final state and ask which methods could produce it.	Knowing which method to use for each task requires practice. Sort for ordering count for frequency index for position and extend for merging are the key tools.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 3 · our school list methods quick reference ·

Differentiation **App** Class 8 Foundation: Focus on sort reverse and count methods with simple lists. Use… **On** Class 9 Application: Use all list methods independently to manage real data. Combine… **Adv** Class 10 Mastery: Use list methods in complex data processing pipelines. Compare sort… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a list of your test scores from the last five tests. Spiral: also recall one fact from last week.

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LESSON 2.03 · LAB · 40 MINUTES

Dictionaries

I CAN Create a dictionary with key-value pairs and understand how keys work · Access modify add and delete dictionary entries using key names · Iterate through dictionaries using keys values and items methods ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students in a dictionary you look up a word to find its meaning. In Python a dictionary works similarly but instead of words and meanings it stores any key and its associated value. Think of it like a school register. The roll number is the key and the student name is the value. You look up by roll number to find the name. Today we learn Python dictionaries which are powerful data structures.	So a dictionary maps keys to values. I can look up any key instantly without searching through the entire collection. That is much faster than a list where I have to check each element one by one.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a dictionary of school houses: houses equals opening curly brace Arya colon green Bharati colon blue Chandrika colon red Drona colon yellow closing curly brace. Now I access a value using houses square bracket Arya which gives green. I can add a new house with houses Ekalavya equals purple. I can change a color with houses Drona equals orange. Try creating your own dictionary and performing these operations.	The curly braces create the dictionary. Keys go on the left of the colon and values on the right. Accessing by key is instant. Adding a new key value pair is simple. Changing a value is like updating an entry. Dictionaries feel more organized than lists.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	A dictionary stores data as key-value pairs. Keys must be unique and immutable like strings numbers or tuples. Values can be any data type. You create a dictionary using curly braces with key colon value pairs separated by commas. Access values using square brackets with the key name. Add new entries by assigning a value to a new key. Modify entries by reassigning. Delete using del statement or pop method. The keys method returns all keys. The values method returns all values. The items method returns key-value pairs as tuples. Dictionaries are unordered in Python versions before 3.7 and ordered by insertion in Python 3.7 and later.	Keys are unique identifiers and values are the data associated with them. Curly braces create dictionaries. Square brackets access values. del and pop remove entries. keys values and items methods help iterate through the data. Dictionaries are perfect for modeling real-world relationships like student records or product catalogs.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a student profile dictionary with keys for name class roll_number subjects and marks. The subjects key should hold a list of subject names. The marks key should hold a list of corresponding marks. Access and print the student name. Access and print the third	My dictionary has name Arya, class 9, roll 15, subjects list with 5 subjects and marks list with 5 marks. I accessed name directly. The third subject is at subjects list index 2. I added	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	subject. Add a new key called attendance with a value of 95. Delete the roll_number key. Print all keys and values using the items method.	attendance 95 and deleted roll_number. The items method printed all key value pairs neatly.	
Evaluate *5 min*	I will give you a scenario and ask you to design a dictionary that models it. You must choose appropriate keys and values and write code to access modify and iterate through the dictionary.	Choosing good keys is important. They should be descriptive and unique. Understanding when to use dictionaries versus lists is a key skill. Dictionaries are best when you need to look up data by a meaningful identifier.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 5 · our school dictionary practice sheet ·

Differentiation **App** Class 8 Foundation: Create simple dictionaries with string keys and values. Practice… **On** Class 9 Application: Create dictionaries with mixed value types including lists. Iterate… **Adv** Class 10 Mastery: Use dictionaries for complex data modeling. Explore nested… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a dictionary representing your daily schedule with time slots as keys and activities as values. Spiral: also recall one fact from last week.

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Lesson 2.04 · lab · 40 minutes

Dictionary Operations

I can Use get method to safely access dictionary values with default values · Check if a key exists using the in operator before accessing it · Merge dictionaries using update method and understand dictionary comprehensions ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students what happens if you try to access a key that does not exist in a dictionary. Python gives a KeyError. Today we learn safe dictionary operations that prevent errors and make our programs robust. Think of it like this: instead of just opening a box and hoping something is inside you check first whether the box contains what you need.	So I can check if a key exists before trying to access it. That prevents my program from crashing. The get method sounds useful because it provides a fallback value if the key is missing.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me try accessing a missing key: student equals name colon Priya. If I try student square bracket age it gives KeyError. But if I use student dot get age 15 it returns 15 the default value. If I use student dot get age it returns None. I can also check with if age in student which returns True or False. Try dictionary comprehension: squares equals bracket key colon key times key for key in range 1, 6 bracket.	The get method prevented the KeyError by returning my default value 15. The in operator checked if age exists as a key. Dictionary comprehension created a new dictionary of squares in one line. These operations make dictionaries safer and more powerful.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The get method provides safe dictionary access. It takes a key and an optional default value. If the key exists it returns the value. If not it returns the default or None. The in operator checks key existence and returns True or False. Always check before accessing to avoid KeyErrors. The update method merges another dictionary into the existing one. If keys overlap the new values replace old ones. Dictionary comprehension creates dictionaries from expressions using the pattern bracket key colon value for item in iterable bracket. These operations make dictionary handling safe and efficient.	Get is the safe way to access values. In checks existence. Update merges dictionaries. Comprehension builds them concisely. Together these tools make dictionaries robust data structures for any program.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Write a program that tracks our school student attendance for a week. Use a dictionary with student names as keys and lists of attendance records as values. Use get to safely access attendance when a new student is added. Check if a student exists before recording attendance. Use update to merge attendance from two sections. Use dictionary comprehension to create a summary dictionary showing total present days for each student.	My attendance dictionary has student names and their daily records. get returns 0 for new students. I check with if name in attendance before recording. update merged Section A and Section B attendance. My comprehension creates total_present for each student by counting present entries in their list.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a program that accesses dictionary keys without safety checks. You must add proper error handling using get and in operator. Then I will ask you to merge two dictionaries and explain what happens with duplicate keys.	Safe dictionary access prevents runtime errors. The get method with a default value is the most practical approach. update replaces old values with new ones when keys conflict.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 7 · our school dictionary operations guide ·

Differentiation **App** Class 8 Foundation: Focus on get method and in operator for safe access. Use predefined… **On** Class 9 Application: Use all dictionary operations independently. Create programs that… **Adv** Class 10 Mastery: Combine dictionary operations with list operations for complex data… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a dictionary of your monthly expenses with categories as keys and amounts as values. Spiral: also recall one fact from last week.

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Lesson 2.05 · lab · 40 minutes

Tuples

I can Create tuples and understand that they are immutable and cannot be changed · Access tuple elements using indexing and slicing similar to lists · Use tuples as dictionary keys and for function multiple return values ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students imagine you write your exam paper in pen. Once written you cannot erase or change it. Tuples in Python work like that. They are like lists but you cannot change them after creation. Think of a tuple as a sealed container. You can look inside but you cannot add remove or change anything. This immutability is actually useful when you want to protect data from accidental changes.	So tuples are permanent while lists are changeable. If I have data that should never change like dates or coordinates a tuple is the right choice. It protects the data from accidental modification.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a tuple: coordinates equals 10 comma 20. I can access with coordinates square bracket 0 which gives 10. I can slice with coordinates 0 colon 2. But if I try coordinates square bracket 0 equals 50 Python gives TypeError. Tuples also work as dictionary keys: location equals bracket coordinates colon school bracket. Try creating your own tuples and see what operations work and which do not.	Indexing and slicing work on tuples just like lists. But assignment does not work because tuples are immutable. I can use tuples as dictionary keys because they are immutable. Lists cannot be dictionary keys because they can change. This is an important difference between tuples and lists.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Tuples are ordered collections enclosed in parentheses or just commas. They are immutable meaning once created they cannot be modified. You cannot add remove or change elements. However you can access elements using indexing and slicing. Tuples can be concatenated using the plus operator to create new tuples. The tuple function converts other iterables to tuples. Tuples are faster than lists and use less memory. They can be used as dictionary keys because they are hashable. Functions can return multiple values as tuples. Unpacking assigns tuple elements to multiple variables at once.	Tuples use parentheses and are immutable. They support indexing slicing and concatenation but not modification. They are faster than lists and can be dictionary keys. Multiple return values from functions come as tuples. Unpacking lets me assign multiple variables from a tuple in one line.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a tuple representing the residential school coordinates with latitude and longitude. Use it as a dictionary key to store the school name. Write a function that returns the student name class and roll number as a tuple. Unpack the returned tuple into three variables. Create a list of tuples where each tuple contains a student name and	My coordinates tuple 12.9716 comma 77.5946 is the key for school name our school. My function returns the tuple with three values. Unpacking gives me name class and roll in separate variables. My student list has tuples sorted by marks using a lambda function. The immutable nature of tuples makes the data safe.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	their marks. Sort this list by marks using the second element of each tuple.		
Evaluate *5 min*	I will give you scenarios and ask whether a tuple or list is more appropriate. You must justify your choice. I will also ask you to write code that uses tuple unpacking and returns multiple values from a function.	Choosing between tuple and list depends on whether the data needs to change. Fixed data like coordinates dates and configurations use tuples. Changeable data like student records and shopping lists use lists.	*Individual exit ticket (2 Q) at door*

Resources

Differentiation **App** Class 8 Foundation: Create simple tuples and practice indexing. Understand that tuples... **On** Class 9 Application: Use tuples as dictionary keys and for multiple return values.... **Adv** Class 10 Mastery: Use tuples in complex data structures. Explore named tuples and... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a tuple of your favorite foods. Spiral: also recall one fact from last week.

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LESSON 2.06 · LAB · 40 MINUTES

Sets

I CAN Create sets and understand they store unique unordered elements · Perform set operations including union intersection and difference · Use sets to remove duplicates from a list and test membership efficiently ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students imagine you have a bag of marbles. A set in Python is like a bag that only allows one marble of each color. If you try to put two red marbles in the bag only one stays. Sets automatically remove duplicates. They are also useful for comparing groups like finding students who are in both the cricket team and the chess team or students who are in cricket but not chess.	So sets automatically remove duplicates. If I have a list with repeated elements converting it to a set gives me only unique elements. Set operations like union and intersection help me compare groups. This is useful for comparing attendance lists or team rosters.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a set of cricket players: cricket equals curly brace Arjun Ravi Kavya Meena closing curly brace. And chess players: chess equals curly brace Ravi Priya Sanjay Arjun closing curly brace. Cricket union chess gives all players from both teams. Cricket intersection chess gives players in both teams like Arjun and Ravi. Cricket difference chess gives players only in cricket. Try creating your own sets and performing these operations.	Union combines both sets without duplicates. Intersection gives only common elements. Difference gives elements in the first set but not the second. These operations are like comparing lists of names to find overlaps and differences. Much faster than writing loops to compare.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Sets are unordered collections of unique elements enclosed in curly braces. Duplicates are automatically removed. You cannot access elements by index because sets are unordered. The add method adds a single element. The remove method deletes an element. The union method combines two sets. The intersection method finds common elements. The difference method finds elements in one set but not the other. The symmetric_difference method finds elements in either set but not both. Sets are useful for removing duplicates and performing membership testing efficiently.	Sets are curly brace collections that enforce uniqueness. No indexing because they are unordered. Union intersection and difference are powerful comparison tools. Adding and removing elements modifies the set. Converting a list to a set removes duplicates. Sets are fast for membership testing.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write a program that finds duplicate entries in a student attendance list. Create a list with some repeated names. Convert it to a set to see unique names. Use set difference to find students who were absent today compared to yesterday. Use intersection to find students present	My attendance list had Arjun appearing three times. Converting to a set gave unique names. Set difference found 2 students absent today but present yesterday. Intersection found 8 students present both days. My subject set shows that Mathematics and English are common across all three classes.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	on both days. Create a set of all subject names across three classes to find common subjects.		
Evaluate *5 min*	I will give you two lists of data and ask you to find common elements unique elements and elements only in one list. You must write the correct set operations and explain what each result represents.	Set operations are efficient for comparing groups. Union gives everyone. Intersection gives common members. Difference gives unique members. Symmetric difference gives members in either but not both.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 9 · our school tuples and lists comparison chart ·

Differentiation **App** Class 8 Foundation: Create sets and practice add and remove. Understand uniqueness... **On** Class 9 Application: Perform all set operations independently. Apply sets to real... **Adv** Class 10 Mastery: Use sets for efficient data analysis. Explore set comprehensions.... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a set of your hobbies and a set of your siblings hobbies. Spiral: also recall one fact from last week.

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LESSON 2.07 · LAB · 40 MINUTES

Nested Structures

I CAN Create lists of lists to represent 2D data like tables and matrices · Create dictionaries with list values and lists with dictionary values · Access and modify elements in nested data structures using multiple indices ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students in mathematics you work with matrices which are grids of numbers. In Python we can represent matrices using nested lists. A list inside a list creates a 2D structure. Think of it like this: a classroom has rows of desks and each desk has items. The outer list is the rows and the inner lists are the desks. Nested structures let us model complex real-world data.	So I can put lists inside lists to create tables. Each inner list represents one row. I can also put lists inside dictionaries to create detailed records. This lets me model complex data like a class register with names and marks.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a 3 by 3 matrix: matrix equals bracket bracket 1 comma 2 comma 3 bracket comma bracket 4 comma 5 comma 6 bracket comma bracket 7 comma 8 comma 9 bracket bracket. Access the center element with matrix square bracket 1 square bracket 1 which gives 5. Now create a student record: student equals bracket name colon Arya comma marks bracket colon bracket 85 comma 90 comma 78 bracket bracket. Access the second mark with student marks bracket 1.	Two indices access elements in nested lists. The first index selects the row and the second selects the column. For dictionaries with list values I access the key first then index into the list. Nested structures make it easy to organize complex related data.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Nested structures are data structures that contain other data structures. A list of lists creates a 2D grid where each inner list is a row. Access uses multiple indices like matrix row column. Dictionaries can contain lists as values allowing one key to hold multiple items. Lists can contain dictionaries allowing each element to be a complex record. Nested dictionaries have dictionaries inside dictionaries. These structures model real-world data like tables student records and organizational hierarchies. Accessing deeply nested data requires multiple brackets or key lookups.	Nested structures combine different data types. Lists of lists create grids. Dictionaries with lists store multiple values per key. Lists of dictionaries store multiple records. Multiple indices or key lookups access deep elements. These structures are essential for modeling complex real-world data.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a residential classroom register as a list of dictionaries. Each dictionary represents a student with keys for name roll_number and marks. The marks key holds a list of marks in different subjects. Write code to find the student with the highest total marks. Calculate the	My register has 10 student dictionaries each with name roll and marks list. I calculated totals by summing each student marks list and found the highest. Subject averages require accessing marks index for each student. Students below 50 are	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	class average for each subject. Find students who scored below 50 in any subject. Add a new student record to the register.	found by checking each mark. Adding a new student is appending a dictionary to the list.	
Evaluate *5 min*	I will give you a nested data structure and ask you to access specific deep elements. You must also write code to create a nested structure from a description and modify specific elements within it.	Accessing nested structures requires understanding the depth. Each index or key peels one layer. Practice tracing through nested access to build confidence with complex data structures.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 11 · our school sets operations visual guide ·

Differentiation **App** Class 8 Foundation: Work with simple nested lists like 2D matrices. Use visual grid... **On** Class 9 Application: Create and manipulate nested structures independently. Work with... **Adv** Class 10 Mastery: Design complex nested structures for real applications. Explore deeply... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a nested list representing the monthly attendance record for your class. Spiral: also recall one fact from last week.

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Lesson 2.08 · lab · 40 minutes

Sorting Algorithms

I can Implement bubble sort by repeatedly comparing and swapping adjacent elements · Implement selection sort by finding the minimum and placing it at the beginning · Compare the efficiency of different sorting algorithms using step counting ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students how does your teacher arrange you in order of height for the assembly photo. The teacher looks at the first two students and swaps them if they are in wrong order. Then looks at the next pair and so on. This is exactly how bubble sort works. Today we learn sorting algorithms which are step-by-step procedures for arranging data in order. These are fundamental algorithms in computer science.	Sorting algorithms are systematic ways to order data. Bubble sort compares neighbors and swaps them. Selection sort finds the smallest and puts it first. I can see how different approaches produce the same sorted result but with different amounts of work.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me demonstrate bubble sort with marks 65 42 78 35 90. Pass 1: compare 65 and 42 swap them. Compare 65 and 78 no swap. Compare 78 and 35 swap. Compare 78 and 90 no swap. After pass 1 the largest value 90 is at the end. Pass 2 continues with remaining elements. Draw this on paper and count the number of comparisons and swaps in each pass.	I drew each pass on paper. Pass 1 needed 4 comparisons and 2 swaps. Pass 2 needed 3 comparisons and 1 swap. Pass 3 needed 2 comparisons and 1 swap. Pass 4 needed 1 comparison and 0 swaps. The list is now sorted. Bubble sort is simple but does many comparisons.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Bubble sort works by repeatedly stepping through the list comparing adjacent elements and swapping them if they are in wrong order. The pass through the list is repeated until no swaps are needed. The largest unsorted element bubbles to its correct position after each pass. Selection sort works by dividing the list into sorted and unsorted portions. It finds the minimum from the unsorted portion and swaps it with the first unsorted element. Both algorithms have quadratic time complexity meaning they get much slower as the list grows. Understanding these algorithms builds the foundation for more efficient ones.	Bubble sort swaps neighbors repeatedly. Selection sort finds minimums and places them. Both are simple but slow for large datasets. Bubble sort does about n squared comparisons. Selection sort also does n squared but fewer swaps. More efficient algorithms exist but these build the foundation.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Implement both bubble sort and selection sort in Python. Create a list of 10 random numbers and sort it using each algorithm. Count the total number of comparisons and swaps for each. Compare the results. Then use Python built-in sort method and note how much faster it is. Discuss why built-in sort is better for real applications.	My bubble sort did 45 comparisons and 12 swaps. My selection sort did 45 comparisons but only 8 swaps. Both produced the same sorted list. The built-in sort was nearly instant because it uses a highly optimized algorithm called	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
		Timsort. Learning the basic algorithms helps understand why efficient sorting matters.	
Evaluate *5 min*	I will give you an unsorted list and ask you to trace through bubble sort or selection sort step by step. You must show each comparison and swap. Then I will ask you to count total operations and compare algorithm efficiency.	Tracing sorting algorithms requires patience. I write the list state after each comparison. Counting operations helps compare algorithms. The key insight is that more efficient algorithms do less work for the same result.	*Individual exit ticket (2 Q) at door*

Resources: Python IDLE · Textbook Chapter 2 Page 13 · our school nested structure examples ·

Differentiation: **App** Class 8 Foundation: Trace bubble sort with small lists of 5 elements. Use visual… **On** Class 9 Application: Implement both bubble sort and selection sort. Count operations and… **Adv** Class 10 Mastery: Analyze time complexity of both algorithms. Explore optimizations like… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write bubble sort to sort your test scores in ascending order. Spiral: also recall one fact from last week.

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Lesson 2.09 · lab · 40 minutes

Searching Algorithms

I can Implement linear search to find an element by checking each position sequentially · Implement binary search to find an element by repeatedly halving the search space · Compare when to use linear search versus binary search based on data properties ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students imagine you are looking for your friend in a crowded school ground. One approach is to check every person one by one. That is linear search. Another approach is to ask half the people if your friend is on the left or right side and then check only that half. That is binary search. Today we learn these two fundamental searching algorithms.	Linear search checks everything one by one. Binary search eliminates half the possibilities each step. Binary search sounds much faster but it requires the data to be sorted first. I can see how the sorted requirement is important.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me demonstrate linear search. I have a list of 8 student roll numbers. I search for roll number 6 by checking each element from the beginning. It takes 6 comparisons. Now for binary search the list must be sorted. I check the middle element. If my target is smaller I search the left half. If larger I search the right half. In a sorted list of 8 elements binary search needs at most 3 comparisons. Try both methods on paper.	Linear search checked 6 elements to find 6. Binary search found it in 3 steps. The sorted list allowed binary search to eliminate half each time. For 8 elements log base 2 of 8 is 3 which matches the binary search steps. Much more efficient than checking all 8.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Linear search checks each element one by one from the beginning. It works on any list whether sorted or not. It takes on average n divided by 2 comparisons and in the worst case n comparisons. Binary search works only on sorted lists. It checks the middle element and eliminates half the remaining elements each step. It takes at most log base 2 of n comparisons. For 1000 elements linear search needs about 500 comparisons while binary search needs about 10. Binary search is dramatically faster for large datasets but requires sorted data.	Linear search is simple but slow. Binary search is fast but needs sorted data. The choice depends on whether the data is sorted and how large the dataset is. Understanding both algorithms helps choose the right tool for each situation.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a residential student database as a sorted list of roll numbers. Implement both linear and binary search to find a specific student. Count the number of comparisons for each algorithm. Test with different target values including the first element last element and middle element. Discuss which algorithm performs better in each case. Create an unsorted list and explain why binary search cannot be used directly.	My linear search found roll 25 in 5 comparisons. Binary search found it in 3 comparisons. For the first element linear search did 1 comparison while binary search still did 3. For unsorted data binary search gives wrong results because it assumes the data is ordered.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a dataset and a search target. You must decide which algorithm to use and justify your choice. Then implement the chosen algorithm and count the comparisons made.	The decision depends on two factors: is the data sorted and how large is it. Sorted data with binary search is always faster for large datasets. Unsorted data requires linear search unless we sort first.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 15 · our school sorting algorithm tracing sheets ·

Differentiation **App** Class 8 Foundation: Focus on linear search implementation and tracing. Use small lists... **On** Class 9 Application: Implement both algorithms and compare their performance. Understand... **Adv** Class 10 Mastery: Analyze time complexity of both algorithms. Explore how database... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a sorted list of your classmates heights. Spiral: also recall one fact from last week.

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LESSON 2.10 · LAB · 40 MINUTES

Stacks and Queues

I CAN Understand stack as a last-in-first-out data structure with push and pop operations · Understand queue as a first-in-first-out data structure with enqueue and dequeue · Apply stacks and queues to solve real-world problems like undo and printing ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students imagine you are stacking plates in a canteen. You put the new plate on top and take the top plate when you need one. This is a stack. Now imagine students lining up at the water fountain. The first person in line gets water first. This is a queue. Today we learn these two important data structures that model many real-world situations.	Stack is last-in-first-out like plates. Queue is first-in-first-out like a line. Both are used everywhere in computing. The undo button uses a stack and the print queue uses a queue.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me implement a stack using a Python list. Push adds to the end with append. Pop removes from the end. Let me push three items: books. Then pop one. The last book pushed is the first one removed. Now a queue: enqueue adds to the end. Dequeue removes from the front using pop index 0. Let me add three names to a queue and remove them. The first name added is the first removed.	Stack push and pop both happen at the end. Queue enqueue happens at the end but dequeue happens at the front. Stack is like undoing actions in reverse order. Queue is like serving customers in the order they arrived.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A stack follows last-in-first-out principle. The last element added is the first one removed. Push adds an element to the top. Pop removes the top element. Peek or top views the top element without removing it. A queue follows first-in-first-out principle. The first element added is the first one removed. Enqueue adds to the back. Dequeue removes from the front. Stacks are used for undo operations function calls and expression evaluation. Queues are used for print jobs task scheduling and breadth-first search.	Stack is LIFO push pop from top. Queue is FIFO enqueue from back dequeue from front. Stacks handle undo and function calls. Queues handle scheduling and processing order. Both are fundamental in computer science.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Implement a stack-based undo system for a simple text editor. The stack stores each text change. Undo pops the last change. Then implement a printer queue that stores print jobs and processes them in order. Add a feature to view the queue without removing items. Discuss how your browser back button works like a stack and how a hospital waiting room works like a queue.	My text editor stack stores each version of the text. Undo pops and restores the previous version. The printer queue adds jobs at the back and prints from the front. My queue view function shows all pending jobs. The browser back button pops from a history stack. The hospital waiting room enqueues patients and dequeues when the doctor is ready.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a real-world scenario and ask you to identify whether it models a stack or queue. You must justify your choice and implement the appropriate data structure.	Stacks are for last-first scenarios like undo and browsing history. Queues are for first-first scenarios like lines and print queues. The key question is whether the most recent or oldest item should be processed first.	*Individual exit ticket (2 Q) at door*

Resources: Python IDLE · Textbook Chapter 2 Page 17 · our school searching algorithm comparison chart ·

Differentiation: **App** Class 8 Foundation: Focus on understanding the concepts of LIFO and FIFO. Use physical… **On** Class 9 Application: Implement stack and queue using lists. Apply them to real scenarios… **Adv** Class 10 Mastery: Implement stack and queue with optimal performance. Explore… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Implement a stack to track your study tasks. Spiral: also recall one fact from last week.

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LESSON 2.11 · LAB · 40 MINUTES

Data Structure Selection

I CAN Compare lists dictionaries sets and tuples to choose the right structure for a task · Analyze the strengths and limitations of each data structure for different scenarios · Apply decision criteria to select the most efficient data structure for a problem ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students you have learned many data structures: lists dictionaries sets tuples stacks and queues. Each has strengths and weaknesses. Today we learn how to choose the right tool for the job. Think of it like a toolbox. You would not use a hammer to tighten a screw. Each tool has a specific purpose. Knowing which tool to use makes you an efficient programmer.	Choosing the right data structure is like choosing the right tool. Lists are flexible dictionaries are fast for lookups sets handle uniqueness and tuples protect data. I need to think about what my data needs before choosing.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me give you three scenarios. Scenario 1: store names of students in a class. Scenario 2: store student names and their marks. Scenario 3: store unique hobby names across the class. For each scenario discuss with your partner which data structure is best and why. Then I will show you the answer and explanation.	Scenario 1 needs a list of names since order matters and duplicates are allowed. Scenario 2 needs a dictionary mapping names to marks for fast lookup. Scenario 3 needs a set because we want unique hobby names without duplicates. Each scenario matches a different data structure.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Lists are ordered and mutable. Use them when you need to maintain insertion order and modify elements. Dictionaries map keys to values. Use them when you need fast lookup by a unique identifier. Sets store unique elements. Use them when you need to remove duplicates or perform membership testing. Tuples are immutable. Use them when data should not change. The choice depends on: whether order matters whether you need key-value lookup whether uniqueness is required and whether mutability is needed. Consider performance for large datasets as different structures have different speeds for different operations.	Lists for ordered mutable data. Dictionaries for key-value mapping. Sets for unique data. Tuples for fixed data. The decision factors are order mutability lookup needs and uniqueness. Understanding these helps choose the most appropriate structure for any problem.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Given these our school scenarios choose the best data structure and justify: storing the school timetable where each period has a subject, tracking unique visitors to the library today, maintaining a shopping list for the tuck shop, recording GPS coordinates of school buildings, and implementing an undo feature in a text editor. Write the justification and a small code example for each.	Timetable needs a dictionary with period as key and subject as value. Library visitors need a set for uniqueness. Shopping list needs a list for order and modification. GPS coordinates need a tuple because they are fixed. Undo feature needs a stack which is a list used as LIFO. Each choice matches the data requirements.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a complex problem and ask you to identify all data structures needed and justify each choice. You must explain why alternatives would be less suitable.	The key is matching data structure properties to problem requirements. If requirements conflict consider which property is most important. Sometimes combining multiple structures is the best approach.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 Page 19 · our school stacks and queues visual guide ·

Differentiation **App** Class 8 Foundation: Focus on the basic differences between lists dictionaries and sets.... **On** Class 9 Application: Analyze multiple scenarios and justify data structure choices.... **Adv** Class 10 Mastery: Analyze complex problems requiring multiple data structures. Consider... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: For each of your school activities identify the best data structure to represent it. Spiral: also recall one fact from last week.

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LESSON 2.12 · LAB · 40 MINUTES

Data Structure Project

I CAN Design a complete program that uses multiple data structures to solve a problem · Implement lists dictionaries and other structures to manage real data · Document the design choices and explain why each data structure was selected ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students this is our final project for the data structures chapter. You will combine everything you have learned into a complete application. This project requires you to think about which data structure best fits each part of your program. Like an architect who chooses different materials for different parts of a building you will choose different structures for different needs.	I am ready to combine all my knowledge of data structures into one project. I need to plan which parts need lists which need dictionaries and which need sets. The planning phase is as important as the coding phase.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let us design a residential Student Record Management System together. We need to store student profiles using dictionaries with name class and marks. We need to track attendance using a list of dictionaries. We need to find unique subjects using a set. We need to sort students by marks using list operations. Let me outline the functions and data structures needed for each feature.	The student profile is a dictionary for fast lookup by name. Attendance is a list because we add records chronologically. Subjects are a set to ensure uniqueness. Sorting uses the list sort method. Each feature uses the most appropriate data structure.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A successful project requires careful planning before coding. Identify all features your program needs. For each feature determine which data structure best fits the requirements. Design functions that use these data structures effectively. Write pseudocode for each function. Implement one function at a time and test thoroughly. Use comments to explain your design decisions. Document why you chose each data structure. Test edge cases like empty inputs and duplicate entries. A well-designed project demonstrates mastery of both programming concepts and data structure selection.	Planning identifies features. Data structure selection matches tools to tasks. Pseudocode guides implementation. Incremental development catches errors early. Documentation explains design reasoning. Testing ensures reliability. The project brings all data structure knowledge together.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Build the complete our school Student Record Management System. Include functions to add student records view all records search by name sort by marks calculate class statistics and generate a report. Use dictionaries for student profiles lists for attendance records and sets for unique subjects. Save all data to a file and load it when the program	My project has seven functions using three data structure types. Student records are dictionaries for fast search. Attendance is a list for chronological records. Subjects are a set for uniqueness. The program saves to file and loads on startup. Error handling catches all invalid inputs. The output is formatted in a clean table.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
	starts. Include error handling for all operations. Format the output neatly.		
Evaluate *5 min*	Your project will be evaluated on: correct use of multiple data structures with justified choices, proper implementation of all required functions, effective error handling, clean formatted output, and documentation of design decisions. Present your project explaining each data structure choice.	My project uses all three main data structures with clear justification for each choice. All functions work correctly. Error handling prevents crashes. Output is professionally formatted. My documentation explains every design decision.	*Individual exit ticket (2 Q) at door*

Resources: Python IDLE · Textbook Chapter 2 Page 21 · our school data structure comparison poster ·

Differentiation: **App** Class 8 Foundation: Complete a simplified project with three functions using lists and… **On** Class 9 Application: Build the complete project with all required functions. Use all… **Adv** Class 10 Mastery: Extend the project with additional features like advanced search and… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a School Inventory Management System. Spiral: also recall one fact from last week.

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LESSON 3.01 · LAB · 40 MINUTES

What is a Database?

I CAN Analyze a database as an organized collection of structured data stored electronically and justify your choice · Analyze and compare real-world examples of databases used in daily life at our school and beyond · Compare file-based storage with database storage explaining advantages of databases ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students imagine you have 500 student records stored in separate text files. To find a specific student you would need to open each file and search. Now imagine all records in one organized system where you can search instantly. That is what a database does. A database is like a well-organized filing cabinet where everything has a specific place and you can find anything quickly.	So a database is a central place to store all related data. Instead of many separate files I have one organized system. Searching and updating data would be much faster and easier.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me show you a real example. Our school library has a system that tracks which student borrowed which book and when it is due. Think about your phone contacts list. Each contact has a name phone number and email. These are examples of databases. Now let me show you the difference. A file-based system stores data in separate text files. A database stores data in tables that are linked together. Try to list five examples of databases you use in your daily life.	My examples are phone contacts school attendance system library catalog online shopping accounts and social media profiles. All of these store data in organized ways. A database is better than files because you can search update and connect data much more efficiently.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A database is an organized collection of structured data stored electronically in a computer system. The word database comes from data meaning facts and base meaning foundation. A database management system or DBMS is software that creates and manages databases. Common DBMS include MySQL PostgreSQL and SQLite. Databases store data in tables with rows and columns. Tables can be linked to each other using relationships. Databases provide fast searching efficient storage data integrity and concurrent access. They are used in schools hospitals banks shops and virtually every organization.	Database is organized data. DBMS is the software that manages it. Tables store data in rows and columns. Relationships connect tables. Databases are faster more organized and more reliable than simple files. They are everywhere in modern life.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Think about residential school systems. Identify at least four different databases the school might use: student records attendance library inventory and exam results. For each database describe what data it stores and who uses it. Then compare how this data would be stored in plain text files versus a database. Discuss the advantages of using a database for each case.	Student records database stores names addresses and parent info used by office staff. Attendance database tracks daily presence used by teachers. Library database tracks books and borrowers used by librarian. Exam results database stores marks used by exam department. In text files searching would	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
		be slow and updating would risk data loss. Databases provide instant search and safe updates.	
Evaluate *5 min*	I will give you a scenario and ask you to design what data a database would store. You must identify the tables needed the columns in each table and who would use the database. You must also explain why a database is better than file storage for this scenario.	A good database design identifies all the data that needs to be stored organizes it into logical tables and considers who will access the data. The key advantage of databases over files is the ability to search and connect related data efficiently.	*Individual exit ticket (2 Q) at door*

Resources Python IDLE · Textbook Chapter 2 review pages · our school project planning template ·

Differentiation **App** Class 8 Foundation: Focus on understanding what a database is through everyday examples.... **On** Class 9 Application: Analyze multiple database examples and their components. Design... **Adv** Class 10 Mastery: Evaluate database design decisions and understand relationships... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write down all the data your school stores about you as a student. Spiral: also recall one fact from last week.

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LESSON 3.02 · LAB · 40 MINUTES

Tables and Records

I CAN Understand that databases store data in tables with rows called records and columns called fields · Analyze and compare the structure of a table including field names data types and primary keys · Create a simple database table and insert records into it ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students a database table is like a spreadsheet but more powerful. Each row represents one complete entry like one student. Each column represents a piece of information like name or marks. Think of your class attendance sheet. Each row is a student and each column is a day of the week. Today we learn how to create and fill database tables.	A database table is like a structured spreadsheet. Rows are records and columns are fields. Each field has a name and a data type that determines what kind of data it can hold.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me create a simple student table. The table has columns for id name class and marks. The id column holds numbers. The name column holds text. The class column holds text. The marks column holds numbers. I will insert five records one for each of you. Watch how each record fits into a row. Try to identify which column is the primary key and why.	The id column looks like the primary key because each student has a unique id number. No two students can have the same id. The name column holds text values like Arjun and Priya. The marks column holds numbers. Each record fills one complete row in the table.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	A table is the basic storage unit in a relational database. Each table has columns also called fields or attributes. Each column has a name and a data type. Common data types include INTEGER for whole numbers REAL for decimal numbers TEXT for strings and DATE for dates. Each row in a table is called a record or tuple. It represents one complete entry. A primary key is a column that uniquely identifies each record. No two records can have the same primary key value. Tables are related to each other through primary and foreign keys.	Tables have columns and rows. Columns define the data types. Rows contain the actual data. The primary key uniquely identifies each row. Different tables connect through keys. This structure organizes data efficiently and allows relationships between different data sets.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a student table in the database with columns for student_id name class section and address. Insert records for five our school students with realistic data. Then create a subjects table with columns for subject_id subject_name and teacher_name. Insert records for five subjects. Observe how each table stores a different type of information. Identify the primary key for each table.	My student table has student_id as the primary key with unique values for each student. The subjects table has subject_id as the primary key. Each table is independent but stores related information about our school. The student table focuses on student information and the subjects table focuses on academic subjects.	*Peer tick against Success Criteria*
EVALUATE *5 min*	I will give you a scenario and ask you to design the table structure including column names data types and which column should be the primary key. You must also insert some sample records and explain your choices.	Good table design requires choosing appropriate column names selecting correct data types and identifying a suitable primary key. The primary key should be unique and never change for a given record.	*Individual exit ticket (2 Q) at door*

Resources	SQLite or MySQL interface · Textbook Chapter 3 Page 1 · our school database examples handout ·
Differentiation	**App** Class 8 Foundation: Focus on understanding the table structure concept. Use spreadsheet… **On** Class 9 Application: Design and create tables with appropriate data types. Insert… **Adv** Class 10 Mastery: Design complex table structures with multiple data types. Understand… **SEND** Trace code with finger and use visual block cards with teacher support
Dormitory · paper	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Design a table to store your personal library reading records. Spiral: also recall one fact from last week.

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LESSON 3.03 · LAB · 40 MINUTES

SQL SELECT

I CAN Write SELECT statements to retrieve all columns or specific columns from a table · Use SELECT DISTINCT to retrieve unique values and eliminate duplicates · Use column aliases and expressions in SELECT to create computed columns ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students SQL is the language used to talk to databases. SELECT is the most common SQL command. It lets you ask the database for specific information. Think of it like asking a librarian a question. You say I want to find books about science. The librarian searches and gives you the results. SELECT works the same way. You ask the database for data and it returns the results.	So SQL SELECT is like asking a question to the database. I can ask for all the data or just specific columns. I can also ask for unique values only. This is much faster than searching through files manually.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me show you some SELECT statements. SELECT star from students retrieves all columns and all rows. SELECT name class from students retrieves only the name and class columns. SELECT distinct class from students shows only the unique class values without repetition. Try these statements on the student table we created and observe the results.	The star symbol means all columns. Choosing specific columns makes the output cleaner. DISTINCT removed duplicate class names so I see Class 8 Class 9 and Class 10 only once each. SELECT is like filtering what I want to see from the complete data.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	The SELECT statement retrieves data from a database table. SELECT star selects all columns. You can specify column names to select only certain columns. SELECT DISTINCT returns only unique values eliminating duplicates. You can use column aliases with the AS keyword to rename columns in the output. Expressions like marks plus 10 can create computed columns. The FROM clause specifies which table to query. SELECT is the foundation of SQL and is used in almost every database operation.	SELECT retrieves data. Star means all columns. Specific column names select those columns only. DISTINCT removes duplicates. AS renames columns. Expressions create new calculated columns. FROM specifies the source table. SELECT is the most fundamental SQL command.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write SELECT statements for the following our school scenarios: retrieve all student names and marks, retrieve only the unique class values, create a computed column showing marks plus 5 bonus points with an alias called adjusted_marks, and retrieve the first three columns of the student table. Test each statement and verify the output matches your expectations.	My first query returns all names and marks. The DISTINCT query shows three unique classes. The computed column adds 5 to each mark with the alias adjusted_marks. The third query specifies three column names. Each SELECT statement answers a different question about the data.	*Peer tick against Success Criteria*
EVALUATE *5 min*	I will give you a database table and ask you to write SELECT statements that retrieve specific information. You must use correct SQL	Writing SELECT statements requires knowing the table structure. I need to remember column names exactly and use	*Individual exit ticket (2 Q) at door*

Phase	Teacher Activity	Student Activity	Assessment Check
	syntax including proper column names table names and any needed keywords.	correct SQL keywords. Practice helps me write queries faster and more accurately.	

Resources SQLite or MySQL interface · Textbook Chapter 3 Page 3 · our school database creation worksheet ·

Differentiation **App** Class 8 Foundation: Focus on basic SELECT star and SELECT specific columns statements.... **On** Class 9 Application: Write SELECT statements with DISTINCT and column aliases. Create... **Adv** Class 10 Mastery: Write complex SELECT statements with expressions and aliases. Combine... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write SELECT statements for a database of your classmates. Spiral: also recall one fact from last week.

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Lesson 3.04 · lab · 40 minutes

SQL WHERE

I can Use the WHERE clause to filter records based on specific conditions · Apply comparison operators equals not equals greater than less than in WHERE conditions · Combine multiple conditions using AND and OR operators in WHERE clauses ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students SELECT alone retrieves all records. But what if you only want students who scored above 80. The WHERE clause lets you filter records based on conditions. Think of it like this: SELECT is like opening a book and WHERE is like turning to a specific page. You only see what matches your criteria.	So WHERE filters the results. I can ask for students above a certain marks or students in a specific class. The database returns only the records that match my condition.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me show you WHERE in action. SELECT star from students where marks greater than 80 returns only students with marks above 80. SELECT star from students where class equals Class 9 returns only Class 9 students. SELECT star from students where marks greater than 70 and class equals Class 8 returns Class 8 students with marks above 70. Try writing your own WHERE conditions.	WHERE with greater than 80 filtered out students below 80. WHERE with equals found students in a specific class. AND combined two conditions so both must be true. I can also use OR to match either condition. WHERE makes SELECT much more powerful.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The WHERE clause filters records based on conditions. It follows the SELECT and FROM clauses. Comparison operators include equals not equals greater than less than greater than or equal and less than or equal. AND requires all conditions to be true. OR requires at least one condition to be true. NOT negates a condition. Conditions can use numbers text and dates. WHERE is essential for extracting specific information from large databases.	WHERE filters rows based on conditions. Comparison operators test values. AND needs all conditions true. OR needs one condition true. NOT reverses a condition. WHERE narrows down results to exactly what you need.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Write WHERE queries for these our school scenarios: find all students who scored above 90, find all Class 10 students, find students who scored below 50 in Mathematics, find students who are in Class 9 OR scored above 85, and find students who are NOT in Class 8. Test each query and verify the results make sense.	My queries return specific subsets of the student data. The above 90 query finds high achievers. The Class 10 query finds senior students. The below 50 query identifies struggling students. The OR query finds either Class 9 or high scorers. The NOT query excludes Class 8 students.	*Peer tick against Success Criteria*
Evaluate *5 min*	I will give you a filtering requirement in plain English and ask you to write the correct SQL WHERE clause. You must also predict how many records will match before running the query.	Translating English conditions to SQL requires understanding comparison operators and logical connectives. AND narrows results while OR widens them. NOT reverses the condition.	*Individual exit ticket (2 Q) at door*

Resources SQLite or MySQL interface · Textbook Chapter 3 Page 5 · our school SQL SELECT practice sheet ·

DIFFERENTIATION	**App** Class 8 Foundation: Focus on simple WHERE conditions with single comparisons like equals… **On** Class 9 Application: Write WHERE queries with AND and OR operators. Combine multiple… **Adv** Class 10 Mastery: Write complex WHERE queries with multiple AND and OR conditions. Use… **SEND** Trace code with finger and use visual block cards with teacher support
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write WHERE queries to find students from your dormitory who scored above 75, students who are in Class 10, and students who are NOT in your class. Spiral: also recall one fact from…

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Lesson 3.05 · LAB · 40 minutes

SQL ORDER BY

I can Use ORDER BY to sort query results in ascending or descending order · Sort by multiple columns to handle ties in sorting criteria · Combine ORDER BY with WHERE to filter and sort results simultaneously ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students our school results are displayed in order from highest to lowest marks. SQL has a similar capability. The ORDER BY clause sorts your query results. Think of it like arranging books on a shelf. You can arrange them alphabetically by title or by author name or by publication year. ORDER BY lets you choose how to arrange your data.	So ORDER BY sorts the results. I can sort marks from high to low to find the class topper. I can sort names alphabetically for a neat display. I can even sort by multiple columns if needed.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me show you ORDER BY. SELECT star from students order by marks descending shows students from highest marks to lowest. Adding ascending sorts from lowest to highest which is the default. ORDER BY class ascending marks descending first sorts by class then within each class sorts by marks from high to low. Try different sorting combinations and observe how the order changes.	Descending order shows highest marks first which is useful for finding toppers. Sorting by class then marks organizes students by class with each class showing highest marks first. Multiple sort columns handle cases where the first column has duplicate values.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The ORDER BY clause sorts the result set. By default it sorts in ascending order. Use DESC for descending order and ASC for ascending order explicitly. You can sort by multiple columns separated by commas. The first column is the primary sort. If values tie the second column breaks the tie. ORDER BY comes after WHERE in the SQL statement. It sorts the filtered results if WHERE is used. ORDER BY is essential for presenting data in a meaningful order.	ORDER BY sorts results. DESC is descending. ASC is ascending which is default. Multiple columns create hierarchical sorting. ORDER BY comes after WHERE. Sorting organizes data for meaningful presentation and analysis.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Write ORDER BY queries for these our school scenarios: display students sorted by marks from highest to lowest to find the class topper, display students sorted alphabetically by name, display students sorted by class then by marks within each class, and display only Class 9 students sorted by marks descending. Verify each result is correctly sorted.	My first query shows Arjun with 98 marks at the top. The alphabetical sort arranges names from A to Z. The multi-column sort groups by class then sorts within each class. The filtered and sorted query shows only Class 9 students in marks order. ORDER BY makes data presentation meaningful.	*Peer tick against Success Criteria*
Evaluate *5 min*	I will give you a requirement like find the top 5 students and ask you to write the SQL query using ORDER BY. You must determine the correct sort order and any additional clauses needed.	ORDER BY with DESC finds highest values. Combined with LIMIT it gives top results. Understanding sort order is essential for answering ranking questions.	*Individual exit ticket (2 Q) at door*

RESOURCES SQLite or MySQL interface · Textbook Chapter 3 Page 7 · our school SQL WHERE practice queries ·

DIFFERENTIATION **App** Class 8 Foundation: Focus on single column ORDER BY with ascending and descending order.... **On** Class 9 Application: Write ORDER BY queries with multiple columns. Combine ORDER BY with... **Adv** Class 10 Mastery: Use ORDER BY in complex queries combining SELECT WHERE and ORDER BY.... **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write ORDER BY queries to display your classmates sorted by marks from highest to lowest, by name alphabetically, and by roll number. Spiral: also recall one fact from last week.

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Lesson 3.06 · lab · 40 minutes

SQL Aggregate Functions

I can Use COUNT to count the number of records matching a condition · Use SUM AVG MAX and MIN to calculate totals averages and find extremes · Use GROUP BY to aggregate data by categories and HAVING to filter groups ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage 5 min	Students when the teacher calculates class average marks or counts how many students passed the exam they are doing aggregate calculations. SQL has built-in functions for these tasks. COUNT counts records SUM adds up values AVG calculates averages MAX finds the highest and MIN finds the lowest. These functions process many records at once.	So I can calculate the class average total marks and highest score using simple SQL functions. COUNT tells me how many students. These functions save me from calculating manually.	*Vocabulary pre-taught; prior ideas on board*
Explore 10 min	Let me demonstrate aggregate functions. SELECT count star from students counts all students. SELECT count star from students where marks greater than 80 counts students above 80. SELECT sum marks from students gives total marks. SELECT avg marks gives the average. SELECT max marks gives the highest and min marks gives the lowest. Try each function and verify the results manually.	COUNT gave 25 which is the total number of students. COUNT with WHERE gave 12 students above 80. SUM of all marks is 1875. Average is 75. Maximum is 98 and minimum is 35. These functions instantly calculate statistics that would take time to compute manually.	*Circulate; verify role rotation every 7 min*
Explain 10 min	Aggregate functions perform calculations on sets of records. COUNT returns the number of rows. COUNT star counts all rows. COUNT column counts non-null values in that column. SUM adds all numeric values. AVG calculates the arithmetic mean. MAX returns the highest value. MIN returns the lowest value. GROUP BY groups records by a column value before applying aggregate functions. HAVING filters groups after aggregation. These functions are essential for data analysis and reporting.	Aggregate functions summarize data. COUNT counts SUM adds AVG averages MAX finds highest and MIN finds lowest. GROUP BY creates groups for per-group aggregation. HAVING filters groups. Together they provide powerful data analysis capabilities.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate 10 min	Write aggregate queries for our school scenarios: count total students in each class, find the average marks for each class, find the highest marks in each subject, count how many students scored above 80 in each class using GROUP BY and HAVING, and find the difference between highest and lowest marks in each class.	COUNT with GROUP BY shows Class 8 has 8 students Class 9 has 10 and Class 10 has 7. AVG by class shows Class 10 has the highest average. MAX by subject shows Mathematics has the highest individual score. HAVING filters classes with more than 2 students above 80. The range shows spread of marks in each class.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a data analysis question like which class has the highest average and ask you to write the SQL query. You must choose the correct aggregate function and any needed GROUP BY or HAVING clauses.	Data analysis questions translate directly to aggregate function queries. The key is identifying which function answers the question and whether grouping is needed.	*Individual exit ticket (2 Q) at door*

Resources SQLite or MySQL interface · Textbook Chapter 3 Page 9 · our school SQL ORDER BY practice ·

Differentiation **App** Class 8 Foundation: Focus on basic aggregate functions COUNT SUM AVG MAX and MIN without… **On** Class 9 Application: Use GROUP BY to aggregate by categories. Apply HAVING to filter… **Adv** Class 10 Mastery: Write complex aggregate queries combining GROUP BY HAVING and ORDER… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write aggregate queries to count how many books you have read this month, find your average rating, find your highest rated book, and find your lowest rated book. Spiral: also recall…

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LESSON 3.07 · LAB · 40 MINUTES

SQL JOIN

I CAN Understand why JOIN is needed to combine data from multiple related tables · Write INNER JOIN to retrieve records that have matching values in both tables · Use LEFT JOIN to retrieve all records from the left table even without matches ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students imagine you have one table with student names and another with their marks. To see a student name with their marks you need to connect the two tables. SQL JOIN does exactly this. Think of it like this: student table is like the left hand and marks table is like the right hand. JOIN brings both hands together to give a complete picture.	So JOIN connects related tables. I can combine student information with their marks even though they are stored in separate tables. This is why databases split data into multiple tables and connect them with keys.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me show you INNER JOIN. I have a students table with id and name and a marks table with student_id and marks. SELECT students dot name marks dot marks from students INNER JOIN marks on students dot id equals marks dot student_id. This combines name and marks from both tables. Try a LEFT JOIN and see what additional records appear.	INNER JOIN showed only students who have marks records. LEFT JOIN showed all students including those without marks. The ON clause specifies how the tables are connected. The student_id in marks table links to id in students table.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	JOIN combines rows from two or more tables based on a related column. INNER JOIN returns only rows with matching values in both tables. LEFT JOIN returns all rows from the left table and matching rows from the right table. Non-matching right rows show NULL. RIGHT JOIN returns all rows from the right table and matching left rows. FULL JOIN returns all rows from both tables. The ON clause specifies the join condition using related columns. JOINs are essential for querying relational databases where data is split across tables.	INNER JOIN finds matches. LEFT JOIN keeps all left rows. RIGHT JOIN keeps all right rows. FULL JOIN keeps everything. ON specifies the matching condition. JOINs connect split data back together for complete queries.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a students table and an enrollments table. The enrollments table has student_id and subject_id columns. Write an INNER JOIN to show which students are enrolled in which subjects. Write a LEFT JOIN to show all students even those not yet enrolled. Write a query to find students enrolled in Mathematics specifically.	My INNER JOIN shows 20 enrollment records connecting students to subjects. LEFT JOIN shows 25 students with 5 showing NULL for unenrolled students. The Mathematics filter shows 8 students enrolled in that subject. JOINs reveal relationships between students and their academic choices.	*Peer tick against Success Criteria*
EVALUATE *5 min*	I will give you two related tables and ask you to write JOIN queries that combine specific columns. You must choose the correct JOIN type based on whether you need only matches or all records from one table.	INNER JOIN is for when both tables must have data. LEFT JOIN is for when you need all records from the primary table. The choice depends on what information you need to display.	*Individual exit ticket (2 Q) at door*

RESOURCES SQLite or MySQL interface · Textbook Chapter 3 Page 11 · our school SQL aggregate functions guide ·

DIFFERENTIATION **App** Class 8 Foundation: Focus on INNER JOIN with simple two-table relationships. Use visual… **On** Class 9 Application: Write INNER JOIN and LEFT JOIN queries independently. Understand… **Adv** Class 10 Mastery: Write complex JOIN queries with multiple tables. Combine JOIN with… **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a database with a students table and a grades table. Spiral: also recall one fact from last week.

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LESSON 3.08 · LAB · 40 MINUTES

Database Design

I CAN Analyze and compare the entities and attributes needed for a database from a description · Design a logical database schema with appropriate tables and columns · Create an Entity-Relationship diagram to visualize database structure ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students before building a house an architect creates a blueprint. Similarly before creating a database you need a design plan. Database design determines what tables you need what columns each table has and how tables relate to each other. Good design prevents problems like duplicate data and inconsistent information.	Database design is the planning phase. I need to identify what data to store organize it into tables and plan how tables connect. Good design saves time and prevents data problems later.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me design a residential library database. First identify the main entities: books students and loans. Books have title author and availability. Students have name class and contact. Loans connect a student to a book with borrow and return dates. Now create tables for each entity with appropriate columns. Draw an Entity-Relationship diagram showing how the tables connect.	My library database has three tables. The books table stores book details with book_id as primary key. The students table stores student info with student_id as primary key. The loans table connects them using both IDs as foreign keys. The ER diagram shows rectangles for tables and lines for relationships.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Database design follows a systematic process. First identify the entities which are the main objects to store data about. Then identify the attributes which are the properties of each entity. Next normalize the data to eliminate redundancy. Then create the schema defining tables columns and data types. Finally establish relationships between tables using primary and foreign keys. An Entity-Relationship diagram visualizes the design with rectangles for entities ovals for attributes and diamonds for relationships. Good design ensures data integrity and efficient queries.	Design steps are identify entities determine attributes normalize data create schema and establish relationships. ER diagrams visualize the design. Good design prevents redundancy and ensures data integrity. Planning saves time during implementation.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Design a residential examination database. Identify entities like students subjects exams and results. Determine attributes for each. Create the database schema with appropriate tables and columns. Draw an ER diagram showing relationships. Implement the design by creating tables in the database and inserting sample data. Test your design by writing queries that answer common questions.	My exam database has four tables. Students store student info. Subjects store subject details. Exams store exam information. Results connect students exams and subjects with marks. The ER diagram shows all relationships clearly. My queries successfully retrieve student results and class statistics.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a real-world scenario and ask you to design a complete database. You must identify entities attributes and relationships create an ER diagram and implement the schema.	Good database design requires understanding the problem thoroughly. Entities come from nouns in the description. Attributes come from properties. Relationships come from how entities interact.	*Individual exit ticket (2 Q) at door*

Resources: SQLite or MySQL interface · Textbook Chapter 3 Page 13 · our school SQL JOIN practice ·

Differentiation:
App Class 8 Foundation: Design simple two-table databases with basic relationships. Use...
On Class 9 Application: Design multi-table databases independently. Create ER diagrams and...
Adv Class 10 Mastery: Design complex normalized databases with multiple relationships....
SEND Trace code with finger and use visual block cards with teacher support

Dormitory · paper: Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Design a database for your personal study planner. Spiral: also recall one fact from last week.

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Lesson 3.09 · lab · 40 minutes

Primary and Foreign Keys

I can Analyze primary key as a unique identifier for each record in a table and justify your choice · Analyze foreign key as a column that references the primary key of another table and justify your choice · Understand how keys establish relationships between database tables ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students every student in our school has a unique roll number. No two students share the same roll number. This roll number is like a primary key in a database. Now when the library records which student borrowed which book it uses the roll number to identify the student. The roll number in the library table is a foreign key because it refers to the roll number in the student table. Keys connect tables together.	So a primary key uniquely identifies each record. A foreign key connects one table to another by referencing the primary key. This is how databases link related data across different tables.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me show you keys in action. The students table has student_id as the primary key. Each student has a unique student_id. The enrollments table has student_id and subject_id. Both are foreign keys. student_id refers to students table and subject_id refers to subjects table. Try inserting a record with a duplicate primary key and see the error. Then insert with a foreign key that does not exist in the referenced table.	The duplicate primary key was rejected because each student_id must be unique. The foreign key error happened because I tried to reference a student_id that does not exist in the students table. Keys enforce data integrity by preventing invalid references.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A primary key uniquely identifies each record in a table. It must be unique and cannot be NULL. A table can have only one primary key. A foreign key is a column that references the primary key of another table. It creates a link between the two tables. Foreign keys can have duplicate values and can be NULL. Keys enforce referential integrity meaning you cannot reference non-existent records. They also enable JOIN operations to combine related data. Keys are the foundation of relational databases.	Primary key is unique identifier for each row. Foreign key links to another table primary key. Keys enforce data integrity. They prevent orphan records and enable table relationships. Without keys databases would be disconnected collections of data.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a school database with a teachers table using teacher_id as primary key and a classes table using class_id as primary key. Create a teaches table linking teachers to classes using both IDs as foreign keys. Insert sample data. Test that you cannot insert a foreign key that references a non-existent teacher. Write a JOIN query to show which teacher teaches which class.	My teachers table has three teachers with unique teacher_id. My classes table has five classes. The teaches table links them with foreign keys. I tried inserting a teaches record with teacher_id 99 which does not exist and got a foreign key error. The JOIN query shows Mr. Sharma teaches Mathematics to Class 10.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a scenario and ask you to identify the primary key for each table and the foreign keys needed to connect them. You must also explain why each key choice is appropriate.	Primary keys should be unique identifiers that never change. Foreign keys should reference appropriate primary keys. The key choices determine how tables can be connected and queried.	*Individual exit ticket (2 Q) at door*

Resources SQLite or MySQL interface · Textbook Chapter 3 Page 15 · our school database design template ·

Differentiation **App** Class 8 Foundation: Focus on understanding primary key uniqueness concept. Identify… **On** Class 9 Application: Create tables with appropriate primary and foreign keys. Insert… **Adv** Class 10 Mastery: Design databases with complex key relationships. Understand composite… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a database with a books table and an authors table. Spiral: also recall one fact from last week.

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LESSON 3.10 · LAB · 40 MINUTES

Database Normalization

I CAN Understand normalization as the process of organizing data to reduce redundancy · Analyze and compare data redundancy and update anomalies in poorly designed tables · Apply basic normalization rules to split a large table into smaller related tables ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students imagine a table that stores student name class teacher name and teacher phone for every student. If ten students have the same teacher the teacher name and phone are repeated ten times. This wastes space and causes problems if the teacher phone changes. Normalization fixes this by splitting data into separate tables. It is like organizing a messy room into separate drawers.	Normalization removes redundant data. Instead of repeating teacher information for every student I store it once in a teachers table and reference it with a foreign key. This saves space and prevents inconsistencies.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me show you a denormalized table with columns student_name class teacher_name teacher_phone subject and marks. Notice how teacher_name and teacher_phone repeat for every student in the same class. Now normalize it by creating a students table a teachers table and a marks table. Compare the before and after. Count how many duplicate values were removed.	The original table had 25 rows with teacher info repeated 25 times. After normalization the teachers table has only 3 rows one for each teacher. The students table has 25 rows with just student data. The marks table connects them. We removed 22 duplicate teacher entries saving significant space.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Normalization is the process of organizing a database to reduce data redundancy and improve data integrity. Redundancy means storing the same data in multiple places. It wastes space and causes update anomalies where changing data in one place but not another creates inconsistency. First Normal Form requires each column to contain atomic values. Second Normal Form requires non-key columns to depend on the entire primary key. Third Normal Form requires non-key columns to depend only on the primary key not on other non-key columns. Normalization splits large tables into smaller focused tables connected by keys.	Normalization eliminates redundancy and prevents anomalies. 1NF requires atomic values. 2NF removes partial dependencies. 3NF removes transitive dependencies. The result is smaller focused tables connected by keys. This saves space and improves data integrity.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Take a denormalized our school table containing student name class subject teacher name and teacher qualification. Normalize it into separate tables. Identify the primary keys for each new table. Draw the relationships. Insert sample data into the normalized tables. Verify that queries on the normalized tables produce the same results as the original table.	My normalized design has three tables. Students with student_id and class. Subjects with subject_id and teacher_id. Teachers with teacher_id and qualification. Queries using JOINs produce identical results to the original table but with no duplicate data. The design is cleaner and more maintainable.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will give you a table with redundant data and ask you to normalize it. You must identify the redundancy create appropriate tables and explain how normalization improves the design.	Normalization requires identifying what data repeats and what depends on what. The key insight is that repeating groups of data should be in separate tables connected by foreign keys.	*Individual exit ticket (2 Q) at door*

Resources SQLite or MySQL interface · Textbook Chapter 3 Page 17 · our school keys and relationships diagram ·

Differentiation **App** Class 8 Foundation: Focus on identifying redundancy in tables. Understand why repeating… **On** Class 9 Application: Normalize tables independently applying basic normalization rules.… **Adv** Class 10 Mastery: Apply 1NF 2NF and 3NF systematically. Analyze complex tables for… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at a school timetable and identify any repeated data. Spiral: also recall one fact from last week.

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Lesson 3.11 · LAB · 40 MINUTES

CRUD Operations

I CAN Implement Create operation using INSERT to add new records to a table · Implement Read operation using SELECT to retrieve data from a table · Implement Update and Delete operations to modify and remove records ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students CRUD stands for Create Read Update and Delete. These are the four fundamental operations for managing data in any database. Think of it like managing your diary. Create means writing a new entry. Read means reviewing what you wrote. Update means changing an entry. Delete means removing an entry. Every database application uses these four operations.	CRUD covers all data management tasks. INSERT creates new records. SELECT reads data. UPDATE modifies records. DELETE removes records. These four operations handle everything I need to do with data.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me demonstrate all four CRUD operations. CREATE: INSERT INTO students name class marks VALUES Arjun Class 10 95. READ: SELECT star from students. UPDATE: UPDATE students set marks equals 98 where name equals Arjun. DELETE: DELETE from students where name equals Arjun. Try each operation and verify the results.	INSERT added a new student record. SELECT displayed all records. UPDATE changed Arjun marks from 95 to 98. DELETE removed the record. Each operation affects the table differently. WHERE is important in UPDATE and DELETE to target specific records.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	CRUD represents the four basic database operations. CREATE uses INSERT INTO to add new records. Specify the table columns and values to insert. READ uses SELECT to retrieve data. Use WHERE to filter specific records. UPDATE modifies existing records using UPDATE table SET column equals value WHERE condition. DELETE removes records using DELETE FROM table WHERE condition. Always use WHERE in UPDATE and DELETE to avoid affecting all records. These operations are the foundation of all database applications.	INSERT adds new data. SELECT retrieves data. UPDATE modifies existing data. DELETE removes data. WHERE clause targets specific records in UPDATE and DELETE. Without WHERE all records are affected. CRUD operations manage all database interactions.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Build a complete CRUD application for our school student records. Implement functions to add a new student read all students update a student marks and delete a student record. Include error handling for each operation. Test all four operations and verify the database state after each change.	My CRUD application has four functions. add_student inserts a new record. view_students displays all records. update_marks modifies a specific student. delete_student removes a record. Each function includes error handling. After all operations the database contains the correct records.	*Peer tick against Success Criteria*
Evaluate *5 min*	I will give you a database scenario and ask you to write CRUD operations for specific requirements. You must use correct SQL syntax and include WHERE clauses where needed.	CRUD operations are straightforward but require attention to syntax and WHERE clauses. Always verify the number of records affected by UPDATE and DELETE operations.	*Individual exit ticket (2 Q) at door*

RESOURCES SQLite or MySQL interface · Textbook Chapter 3 Page 19 · our school normalization examples ·

DIFFERENTIATION **App** Class 8 Foundation: Focus on basic INSERT and SELECT operations. Practice adding and… **On** Class 9 Application: Implement all four CRUD operations. Write UPDATE and DELETE with… **Adv** Class 10 Mastery: Implement complete CRUD applications with error handling and… **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a CRUD application for managing your personal contact list. Spiral: also recall one fact from last week.

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LESSON 3.12 · LAB · 40 MINUTES

Database Project

I CAN Design and implement a complete database for a real-world our school scenario · Apply all SQL concepts including CREATE SELECT WHERE JOIN and aggregates · Document the database design and demonstrate querying capabilities ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students this is the final project for our database chapter. You will apply everything you learned about databases from design to implementation to querying. This project requires you to think like a database administrator who designs the schema creates the tables populates them with data and writes queries to extract useful information.	I am ready to combine all my database knowledge into one complete project. I need to plan the database design implement it with SQL and write queries to answer real questions about the data.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let us plan a residential Student Information System together. First identify the main data needs: student profiles attendance records exam results and extracurricular activities. Design tables for each. Consider relationships between tables. Plan the queries needed to answer common questions like who is the class topper and what is the attendance percentage.	My system needs four main tables. Students for basic info. Attendance for daily records. Results for exam marks. Activities for clubs and sports. The queries need JOINs to connect these tables. Aggregate functions will answer statistical questions.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A database project follows the complete database lifecycle. Start with requirements analysis to understand what data to store. Design the schema including tables columns keys and relationships. Implement by creating tables with SQL DDL statements. Populate with sample data using INSERT statements. Query the database to extract information. Test thoroughly to ensure correctness. Document the design and queries for future maintenance. The project demonstrates mastery of all database concepts.	Project lifecycle is design implement populate query test and document. Each phase builds on the previous one. Good design makes implementation easier. Thorough testing ensures correctness. Documentation helps maintenance and future development.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Build the complete our school Student Information System. Create at least four related tables with appropriate primary and foreign keys. Insert at least 20 records across all tables. Write queries using SELECT WHERE JOIN GROUP BY and ORDER BY to answer ten different questions about the data. Include questions about student performance attendance and activity participation.	My system has four tables with 25 student records 50 attendance records and 30 result records. My queries find class averages attendance percentages top performers in each subject and students who participate in activities. The database provides complete information about our school.	*Peer tick against Success Criteria*
Evaluate *5 min*	Your project will be evaluated on: database design quality with appropriate normalization, correct SQL syntax for all operations,	My project meets all criteria. The database is well normalized. All SQL operations work correctly. My queries answer diverse	*Individual exit ticket (2 Q) at door*

Phase	Teacher Activity	Student Activity	Assessment Check
	comprehensive queries answering multiple questions, proper use of keys and relationships, and clear documentation of design decisions.	questions about the data. Keys properly connect the tables. My documentation explains every design decision.	

Resources SQLite or MySQL interface · Textbook Chapter 3 Page 21 · our school CRUD operations reference ·

Differentiation
- **App** Class 8 Foundation: Create a simplified two-table database with basic queries. Follow...
- **On** Class 9 Application: Build the complete four-table database independently. Write diverse...
- **Adv** Class 10 Mastery: Extend the database with additional tables and complex queries. Add...
- **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a Library Management System database. Spiral: also recall one fact from last week.

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LESSON 4.01 · CONCEPT · 40 MINUTES

How the Web Works

I CAN Explain the client-server model and how browsers request web pages · Understand the role of URLs IP addresses and DNS in web navigation · Describe the HTTP request-response cycle that powers web communication ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students when you type a website address in your browser and press Enter something amazing happens in seconds. Your browser sends a request to a computer called a server far away and the server sends back the web page. Today we learn how this client-server communication works. Think of it like ordering food by phone. You are the client calling the restaurant the server. You place an order the request and they deliver food the response.	So my browser is the client and the website is stored on a server. When I visit a website my browser asks the server for the page and the server sends it back. This happens every time I visit any website.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me trace what happens when you visit kreis.edu. First your browser needs the servers address. It asks DNS which is like a phone book for websites. DNS translates kreis.edu to an IP address like 192.168.1.1. Then your browser sends an HTTP request to that address. The server receives the request and sends back an HTTP response containing the HTML code. Your browser reads the HTML and displays the page. Try to draw this process as a flowchart.	I drew the flowchart: browser asks DNS for IP address DNS returns the address browser sends HTTP request to server server processes request server sends HTTP response with HTML browser renders the HTML and displays the page. The whole process takes less than a second.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	The web operates on a client-server model. The client is your web browser like Chrome or Firefox. The server is a computer that stores websites. Communication happens through HTTP or HTTPS which is the language of the web. A URL is the address of a web resource. It contains the protocol domain name and path. DNS translates domain names to IP addresses. The browser sends an HTTP request to the server. The server processes the request and sends an HTTP response containing the web page content. The browser then renders the content for you to see.	Client-server model means browsers request and servers respond. HTTP is the communication language. URLs are web addresses. DNS translates names to numbers. Requests go out and responses come back. This cycle happens for every web page you visit.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a diagram showing the complete web communication process including DNS resolution HTTP request and response and page rendering. Label each component and describe what happens at each step. Then research and explain the difference between HTTP and HTTPS. Why should you always look for HTTPS when entering personal information online.	My diagram shows six steps from typing the URL to seeing the page. DNS is step one translating the name. HTTP request is step two. Server processing is step three. HTTP response is step four. Browser rendering is step five. Display is step six. HTTPS adds encryption making the communication secure.	*Peer tick against Success Criteria*

Phase	Teacher Activity	Student Activity	Assessment Check
Evaluate *5 min*	I will ask you to explain the complete process of visiting a website from typing the URL to seeing the page. You must identify each component and its role in the process.	Understanding the web process requires knowing each component. DNS translates names. HTTP carries requests and responses. Servers store and serve content. Browsers render and display pages.	*Individual exit ticket (2 Q) at door*

Resources SQLite or MySQL interface · Textbook Chapter 3 review pages · our school project requirements document ·

Differentiation **App** Class 8 Foundation: Focus on the basic client-server concept using the restaurant… **On** Class 9 Application: Understand DNS HTTP and the request-response cycle in detail.… **Adv** Class 10 Mastery: Analyze the complete web communication process including HTTP methods… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a diagram showing how a web page travels from the server to your computer. Spiral: also recall one fact from last week.

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LESSON 4.02 · CONCEPT · 40 MINUTES

HTML Basics

I CAN Write a basic HTML document with proper structure including head and body · Use heading paragraph and line break tags to structure content · Open and view HTML files in a web browser to see the rendered output ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students every website you visit is built using HTML which stands for HyperText Markup Language. HTML is the skeleton of a web page. Think of it like this: if a web page were a human body HTML would be the bones that give it structure. CSS would be the skin and clothes that make it look good. JavaScript would be the muscles that make it move. Today we learn HTML which provides the structure.	So HTML is the foundation of every website. It defines what content appears on the page like headings paragraphs and images. Without HTML there would be no web pages to see.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me create your first HTML page. I will write a simple document with a heading and a paragraph. The document starts with doctype html then html tags. Inside html we have head and body sections. The head contains the title. The body contains the visible content. Save this as index dot html and open it in a browser. What do you see.	I see my first web page with a heading Hello our school and a paragraph Welcome to our school website. The browser displayed the heading larger than the paragraph. The title appears in the browser tab. HTML tags tell the browser how to display the content.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	HTML uses tags to mark up content. Tags come in pairs with an opening tag and a closing tag. The closing tag has a forward slash. The head section contains metadata like the title and links to stylesheets. The body section contains the visible page content. Heading tags h1 through h6 create headings with h1 being the largest. The p tag creates paragraphs. The br tag creates line breaks. HTML is not a programming language but a markup language that describes the structure of content.	HTML tags come in pairs. Head holds metadata. Body holds content. Headings use h1 to h6. Paragraphs use p tags. HTML describes structure not behavior. The browser reads HTML and decides how to display the content.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create an HTML page about residential school. Include a main heading h1 with the school name. Add a subheading h2 with About Our School. Write two paragraphs describing the school. Add another subheading h2 with Our Values. List three values in separate paragraphs. Open the page in a browser and verify the structure displays correctly.	My HTML page has the school name as the main heading. About Our School section describes the school in two paragraphs. Our Values section lists three values each in its own paragraph. The browser shows headings in decreasing size and paragraphs with spacing.	*Peer tick against Success Criteria*
Evaluate *5 min*	I will give you a content description and ask you to write the correct HTML structure. You must use appropriate heading levels paragraph tags and proper document structure.	HTML structure follows a hierarchy. h1 is the main heading h2 is section heading and so on. Paragraphs use p tags. The document must have proper html head and body structure.	*Individual exit ticket (2 Q) at door*

RESOURCES Web browser · Textbook Chapter 4 Page 1 · our school web communication diagram handout ·

Differentiation	**App** Class 8 Foundation: Create a simple HTML page with one heading and one paragraph. Use… **On** Class 9 Application: Create HTML pages with multiple heading levels and paragraphs.… **Adv** Class 10 Mastery: Create well-structured HTML documents with semantic organization.… **SEND** Trace code with finger and use visual block cards with teacher support
Dormitory · paper	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create an HTML page about your dormitory. Spiral: also recall one fact from last week.

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Lesson 4.03 · concept · 40 minutes

HTML Elements

I can Use list tags to create ordered and unordered lists on web pages · Add images to HTML pages using the img tag with src and alt attributes · Create hyperlinks using anchor tags to connect web pages together ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students yesterday we created basic text pages. Today we learn HTML elements that make pages more useful and interesting. Lists organize information. Images make pages visual. Links connect pages together. Think of these elements as the furniture in a room. Without furniture a room is empty. Without these elements a web page is just plain text.	Lists images and links are essential web page elements. Lists organize content images add visual interest and links connect pages. Together they make web pages functional and engaging.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me show you lists first. ul creates an unordered bullet list and ol creates an ordered numbered list. Each item uses the li tag. Now images: img src equals photo dot jpg alt equals description creates an image. The alt text appears if the image cannot load. Links: a href equals URL creates a clickable link. Try creating each.	My bullet list shows school subjects with dots. My numbered list shows steps in a process with numbers. My image displays a photo with the alt text showing when I removed the image file. My link clicks through to the school website. These elements bring pages to life.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	HTML elements extend basic text with rich content. The ul element creates unordered lists with bullet points. The ol element creates ordered lists with numbers. The li element defines each list item. The img element embeds images using the src attribute for the image path and alt attribute for alternative text. The a element creates hyperlinks using the href attribute for the destination URL. These elements are fundamental for creating functional web pages.	Lists organize content with ul for bullets and ol for numbers. Images use img with src and alt. Links use a with href. These three element types transform plain text into a rich web page.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create an HTML page about our school with at least three different element types. Include an unordered list of school facilities an ordered list of daily schedule and an image of the school. Add at least two links to other pages or external websites. Use proper alt text for the image and meaningful link text.	My page has a bullet list of facilities including library lab and playground. My numbered list shows the daily schedule. The image shows the school building with alt text. Links connect to the admission page and education department website.	*Peer tick against Success Criteria*
Evaluate *5 min*	I will give you a content description and ask you to write the correct HTML using appropriate elements. You must choose between ordered and unordered lists and provide proper attributes.	Choosing between ul and ol depends on whether order matters. Images always need alt text for accessibility. Links need meaningful text that describes the destination.	*Individual exit ticket (2 Q) at door*

Resources Text editor and web browser · Textbook Chapter 4 Page 3 · our school HTML basics cheat sheet ·

DIFFERENTIATION	**App** Class 8 Foundation: Create simple lists and add one image to a page. Use provided image... **On** Class 9 Application: Create pages with multiple element types including lists images and... **Adv** Class 10 Mastery: Create complex pages with nested lists multiple images and internal... **SEND** Trace code with finger and use visual block cards with teacher support
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create an HTML page about your favorite book. Spiral: also recall one fact from last week.

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LESSON 4.04 · CONCEPT · 40 MINUTES

HTML Forms

I CAN Create HTML forms with input fields for text email and password data · Use form controls like buttons dropdowns and checkboxes for user interaction · Understand the role of forms in collecting user input for web applications ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students when you fill out an online form to create an account or submit homework you are interacting with HTML forms. Forms are how websites collect information from users. Today we learn to create forms. Think of a form like a paper application form but digital. It has fields for you to fill in and a submit button to send your answers.	HTML forms collect user input. I see forms everywhere for login registration and search. Learning to create forms lets me build interactive web pages that can gather information.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me create a student feedback form. The form element wraps all input fields. Input type text creates a text field. Input type email validates email format. Input type password masks the input. Input type submit creates a submit button. A label element connects to an input using the for attribute. Try creating a form with at least three different input types.	My form has a name field with type text an email field with type email and a feedback textarea for longer messages. The submit button says Submit Feedback. When I click submit the form tries to send the data. The email field shows an error if I enter invalid email format.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	HTML forms collect user input. The form element is the container for all form controls. Input elements create different field types: text for single-line text email for email validation password for masked input number for numeric values and checkbox for multiple selections. The textarea element creates multi-line text areas. Select and option elements create dropdown menus. The button element creates clickable buttons. Labels improve accessibility by connecting text descriptions to inputs.	Forms contain input controls. Text email password number and checkbox are input types. Textarea handles multi-line input. Select creates dropdowns. Labels improve accessibility. Submit sends the form data. Forms enable user interaction.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a residential admission inquiry form with fields for student name parent name email phone number class applying for and a textarea for additional questions. Add a dropdown for preferred section and checkboxes for facilities interested in. Include proper labels for each field and a submit button.	My admission form collects all necessary information. Name and parent name use text inputs. Email and phone use appropriate types. Class uses a dropdown with Class 8 9 and 10 options. Facilities uses checkboxes for library lab and sports. The textarea collects additional questions.	*Peer tick against Success Criteria*
EVALUATE *5 min*	I will give you a form description and ask you to write the HTML code. You must choose appropriate input types and include labels for accessibility.	Choosing input types depends on the data. Text for names email for email addresses password for passwords and number for numeric values. Labels are essential for accessibility.	*Individual exit ticket (2 Q) at door*

RESOURCES Text editor and web browser · Textbook Chapter 4 Page 5 · our school HTML elements reference ·

DIFFERENTIATION **App** Class 8 Foundation: Create a simple form with text and email inputs only. Use provided… **On** Class 9 Application: Create forms with multiple input types including text email… **Adv** Class 10 Mastery: Create complex forms with validation attributes. Understand form… **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create an HTML form for a library book request. Spiral: also recall one fact from last week.

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Lesson 4.05 · concept · 40 minutes

CSS Introduction

I can Understand that CSS controls the visual presentation of HTML elements · Apply inline internal and external CSS to style web pages · Use CSS properties to change colors fonts and backgrounds ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students HTML provides the structure of a web page but CSS makes it beautiful. CSS stands for Cascading Style Sheets. Think of HTML as the skeleton and CSS as the skin and clothes. Without CSS every web page would look plain black text on white background. CSS adds color style and visual appeal. Today we learn how to make our web pages look professional.	So CSS is like the makeup and clothing of a web page. HTML defines what content appears and CSS defines how it looks. I can change colors fonts sizes and backgrounds using CSS.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me show you three ways to add CSS. Inline CSS goes directly on an element using the style attribute. Internal CSS goes in the head section using a style tag. External CSS goes in a separate dot css file linked with a link tag. Try changing the text color using each method. Which is easiest for quick changes and which is best for many pages.	Inline is easy for one element. Internal works for one page. External is best for many pages because one CSS file styles all linked pages. External is the most professional approach.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	CSS controls the visual presentation of HTML elements. Inline CSS uses the style attribute on individual elements. Internal CSS uses a style tag in the document head. External CSS uses a separate file linked with the link tag. CSS properties include color for text color background-color for background color font-size for text size font-family for typeface and text-align for horizontal alignment. The selector determines which elements the styles apply to. CSS separates content from presentation.	Three CSS methods: inline on elements internal in head external in separate file. Properties control color font size and alignment. Selectors target elements. External CSS is the best practice for maintainability.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create an external CSS file to style your our school HTML page. Set the body background to light blue. Make h1 headings dark green and centered. Set paragraph text to dark gray with font size 16 pixels. Style links with underline and hover color change. Link the CSS file to your HTML page and verify.	My CSS file sets the background to light blue. Headings are dark green and centered. Paragraphs are dark gray at 16 pixels. Links are blue with underline and change to red on hover. The HTML page loads the CSS file and displays all styles.	*Peer tick against Success Criteria*
Evaluate *5 min*	I will give you an HTML page and ask you to write CSS to achieve a specific visual appearance. You must choose appropriate properties and values.	CSS properties map to visual attributes. Color changes text color. Background-color changes the background. Font-size controls text size. The selector determines which elements receive the styles.	*Individual exit ticket (2 Q) at door*

Resources Text editor and web browser · Textbook Chapter 4 Page 7 · our school HTML forms reference ·

Differentiation
App Class 8 Foundation: Apply inline CSS to change text color and background. Focus on…
On Class 9 Application: Create internal and external CSS files. Apply multiple properties…
Adv Class 10 Mastery: Create comprehensive CSS stylesheets with multiple selectors and…
SEND Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a CSS file to style your HTML dormitory page. Spiral: also recall one fact from last week.

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LESSON 4.06 · CONCEPT · 40 MINUTES

CSS Selectors

I CAN Use element class and ID selectors to target specific HTML elements · Understand selector specificity and how it determines which styles apply · Apply pseudo-class selectors to style elements based on user interaction ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students what if you want to style only some paragraphs differently. CSS selectors tell CSS exactly which elements to style. Think of calling students by name versus calling everyone.	Selectors target specific elements. Element selectors style all of a type. Class selectors style groups. ID selectors style one unique element.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	The element selector p styles all paragraphs. The class selector dot highlight styles elements with that class. The ID selector hash header styles one element. Try adding a class to one paragraph and styling it differently.	My element selector changed all paragraphs. My class selector made only the highlighted paragraph yellow. ID selector made only the header red.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	CSS selectors determine which elements receive styles. Element targets all of a type. Class uses a dot prefix. ID uses a hash prefix. Specificity determines which style wins: ID is more specific than class which is more specific than element. Pseudo-class selectors like hover style elements based on state.	Element targets types. Class targets groups. ID targets one element. Specificity order is ID then class then element. Pseudo-classes target states like hover.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Create a styled our school page using all selector types. Use element selector for body text. Create a class called special for important paragraphs. Use an ID for the main heading. Add hover effects on links.	My page uses element selector for paragraphs the special class for highlighted content the ID for the heading and hover effects on links.	*Peer tick against Success Criteria*
EVALUATE *5 min*	Write CSS using correct selectors to target specific elements without affecting others.	IDs for unique elements classes for groups and elements for all of a type.	*Individual exit ticket (2 Q) at door*

RESOURCES Text editor and web browser · Textbook Chapter 4 Page 9 · our school CSS introduction guide ·

DIFFERENTIATION **App** Class 8 Foundation: Focus on element and class selectors with teacher guidance. **On** Class 9 Application: Use all selector types including ID and pseudo-class. **Adv** Class 10 Mastery: Apply advanced selectors and understand specificity rules. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Style your dormitory page using element class and ID selectors. Spiral: also recall one fact from last week.

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LESSON 4.07 · CONCEPT · 40 MINUTES

CSS Layout

I CAN Use the box model concept including margin border padding and content area · Apply display property values like block inline and flex to control layout · Create multi-column layouts using CSS flexbox for modern web design ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students every HTML element is like a box. Content is inside then padding then border then margin. This is the CSS box model. Think of a gift box. The gift is content wrapping is padding box edge is border and space between boxes is margin.	The box model explains how elements take up space. Padding adds space inside border adds a line and margin adds space outside.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Create two boxes with different padding and margin. Box one has padding 20 and margin 10. Box two has padding 5 and margin 30. Now try flexbox. A container with display flex arranges children in a row. Use justify-content to center items.	Box one has more internal space. Box two has more external space. Flexbox lined up items and justify-content center moved them to the middle.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The box model has content padding border and margin. Content is actual text. Padding is space inside the border. Margin is space outside. Display property controls flow: block takes full width inline flows with text and flex creates flexible layouts. Flexbox provides alignment and distribution tools.	Box model is content plus padding plus border plus margin. Display block is full width. Inline is flowing. Flex creates responsive layouts.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a residential website layout with header sidebar and main content using flexbox. Set header to span full width. Use flex for sidebar and content side by side. Apply padding and margin. Add a footer spanning full width.	My layout has a green header spanning full width. Sidebar takes 25 percent and content takes 75 percent. Footer sits at the bottom.	*Peer tick against Success Criteria*
Evaluate *5 min*	Write CSS to achieve a described page layout using box model and flexbox.	Layout requires the box model for spacing and flexbox for arrangement.	*Individual exit ticket (2 Q) at door*

RESOURCES Text editor and web browser · Textbook Chapter 4 Page 11 · our school CSS selectors reference ·

DIFFERENTIATION **App** Class 8 Foundation: Focus on box model. Practice adjusting padding and margin. **On** Class 9 Application: Use flexbox to create multi-column layouts independently. **Adv** Class 10 Mastery: Create complex responsive layouts using flexbox. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a layout for your dormitory page with header and two columns using flexbox. Spiral: also recall one fact from last week.

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LESSON 4.08 · CONCEPT · 40 MINUTES

Responsive Design

I CAN Understand why web pages need to work on different screen sizes · Use media queries to apply different CSS rules based on screen width · Apply the viewport meta tag and relative units for mobile-friendly design ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students you browse websites on phones tablets and laptops. Each has a different screen size. Responsive design makes pages adapt to any size. Think of water taking the shape of its container.	Responsive design means one website works on all devices. One adaptive site adjusts its layout automatically.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	The viewport meta tag tells the browser to use actual device width. Media queries apply CSS rules based on screen width. Try resizing your browser window and watch the layout change at breakpoints.	The viewport tag made text readable on phone. The media query changed the layout when I made the window narrow. Layout switched from two columns to one.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Responsive design ensures pages work on all devices. Viewport meta tag tells the browser to use actual width. Media queries use syntax like at media max-width 600px. Relative units like percentages create flexible layouts. Mobile-first starts with smallest screen. Breakpoints are widths where layout changes occur.	Viewport enables mobile rendering. Media queries detect screen size. Relative units create flexible sizing. These techniques create adaptive pages.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Make your our school website responsive. Add viewport meta tag. Use media queries to stack columns on screens smaller than 600 pixels. Change font sizes for small screens. Test by resizing browser.	My website now has the viewport tag. Media query stacks sidebar below content on small screens. Font sizes decrease for mobile.	*Peer tick against Success Criteria*
EVALUATE *5 min*	Make a desktop layout responsive by adding viewport tag media queries and adjusting layouts.	Responsive design requires viewport tag media queries and relative units.	*Individual exit ticket (2 Q) at door*

RESOURCES Text editor and web browser · Textbook Chapter 4 Page 13 · our school CSS layout examples ·

DIFFERENTIATION **App** Class 8 Foundation: Add viewport meta tag and observe the difference. **On** Class 9 Application: Create responsive layouts with media queries independently. **Adv** Class 10 Mastery: Design mobile-first responsive websites with breakpoints. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Make your dormitory page responsive with a viewport tag and media query. Spiral: also recall one fact from last week.

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LESSON 4.09 · CONCEPT · 40 MINUTES

JavaScript Introduction

I CAN Understand that JavaScript adds interactivity and behavior to web pages · Write JavaScript variables loops and conditionals for client-side logic · Use the alert function and console to display information and debug code ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students HTML gives structure CSS gives style and JavaScript gives life to web pages. JavaScript makes pages interactive. If HTML is the body and CSS is the clothes then JavaScript is the brain.	JavaScript makes web pages interactive. I can create programs that run in the browser and respond to user actions.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Add JavaScript using a script tag. Alert shows a popup box. Document dot write adds text to the page. Console dot log sends messages to the developer console. Try these and see what happens.	Alert showed a popup. Document dot write added text. Console dot log sent a message I could see by pressing F12.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	JavaScript is a programming language that runs in browsers. Write in script tags or external js files. Variables store data using let and const. Conditionals use if-else. Loops repeat using for and while. Alert shows popups. Console dot log helps debugging.	JavaScript is the behavior layer. Variables store data. Conditionals decide. Loops repeat. Alert shows popups. Console helps debugging.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a JavaScript program that greets the user by name. Use a variable for the name. Use if-else to check if empty and show default greeting. Add a for loop printing 1 to 5 using console dot log.	My program stores the name in a variable. If-else checks for empty name. The for loop prints 1 through 5.	*Peer tick against Success Criteria*
Evaluate *5 min*	Write JavaScript using variables conditionals and loops to solve a problem.	Variables store data. Conditionals decide. Loops repeat.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Write simple JavaScript with alert and document write. **On** Class 9 Application: Write programs with variables conditionals and loops. **Adv** Class 10 Mastery: Create interactive programs with complex logic. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a JavaScript program that calculates your dormitory room area. Spiral: also recall one fact from last week.

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Lesson 4.10 · concept · 40 minutes

DOM Manipulation

I can Understand the Document Object Model as a tree of HTML elements · Use getElementById and querySelector to select HTML elements · Modify element content styles and attributes dynamically ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students the browser builds a tree from HTML called the Document Object Model or DOM. JavaScript can find any element and change it. Think of a family tree where JavaScript can find any member.	The DOM is a map of the web page. JavaScript uses it to find and change elements without reloading the page.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Select an element with document dot getElementById and change its text content. Use querySelector to select by CSS selector. Change style using element dot style dot color equals red. Watch the page change without reloading.	I selected the heading and changed its text and color. The page updated instantly.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The DOM is a tree of HTML elements the browser creates. getElementById finds by ID. querySelector finds by CSS selector. textContent changes text. Style properties change appearance. addEventListener responds to events. The DOM bridges HTML and JavaScript.	DOM is the browser tree. getElementById and querySelector find elements. textContent and style change them. addEventListener makes them respond.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a residential page with a button that changes background color when clicked. Use getElementById to select the button. Add a click event listener. Change document dot body dot style dot backgroundColor on click.	My button changes background from white to blue on click. Each click cycles through colors.	*Peer tick against Success Criteria*
Evaluate *5 min*	Write JavaScript that selects HTML elements and modifies their content or style.	DOM methods find elements. Properties change them. Events trigger changes.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Use getElementById to select and change one element. **On** Class 9 Application: Use querySelector and event listeners for interaction. **Adv** Class 10 Mastery: Create complex DOM manipulation with multiple events. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Add a button that shows and hides your daily routine paragraph. Spiral: also recall one fact from last week.

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Lesson 4.11 · concept · 40 minutes

Web Security

I can Analyze and compare common web threats including phishing and cross-site scripting · Understand HTTPS encryption and why secure connections protect data · Apply safe browsing habits to protect personal information online ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students the web has dangers. Hackers try to steal passwords and personal data. Today we learn about web security threats and protection. Think of it like locking your house door.	Web security means protecting yourself online. I need to recognize threats and know how to stay safe.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let me show you what a phishing email looks like. It tries to trick you into clicking fake links. HTTPS encrypts data between browser and server. Check for the lock icon in the address bar.	The phishing email had suspicious links. HTTPS sites show a lock icon. HTTP sites show not secure warning.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Web security threats include phishing which tricks users into revealing data and malware which damages devices. HTTPS encrypts communication. SSL certificates verify website identity. Strong passwords protect accounts. Two-factor authentication adds extra security.	Phishing tricks users. Malware damages devices. HTTPS encrypts data. SSL verifies identity. Strong passwords protect accounts.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a web security awareness poster for our school. Include tips for creating strong passwords recognizing phishing emails and checking HTTPS. Add a section about social media privacy.	My poster includes password tips recognizing phishing by checking sender addresses and looking for HTTPS lock icon.	*Peer tick against Success Criteria*
Evaluate *5 min*	Analyze and compare security threats in given scenarios and recommend protection measures.	Recognizing threats requires awareness. Protection involves technical and behavioral measures.	*Individual exit ticket (2 Q) at door*

Resources Text editor and web browser · Textbook Chapter 4 Page 19 · our school DOM guide ·

Differentiation **App** Class 8 Foundation: Focus on recognizing phishing and using HTTPS. **On** Class 9 Application: Understand encryption and implement password best practices. **Adv** Class 10 Mastery: Analyze security vulnerabilities and design protection strategies. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a checklist of web security practices to follow every day. Spiral: also recall one fact from last week.

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LESSON 4.12 · CONCEPT · 40 MINUTES

Portfolio Website

I CAN Combine HTML CSS and JavaScript to build a complete personal portfolio website · Apply responsive design principles to ensure the site works on all devices · Present and document the website with design decisions and technical choices ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students this is our final web project. You will combine everything learned about HTML CSS and JavaScript into a personal portfolio website. This is like building your digital identity.	I am ready to combine all web technologies into one complete project. I need to plan structure style and add interactive elements.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Let us plan a residential student portfolio. We need pages for home about skills and contact. HTML provides structure. CSS provides styling. JavaScript adds interactivity like a dark mode toggle.	My portfolio has four sections. HTML structures the content. CSS styles everything. JavaScript toggles dark mode. Media queries make it responsive.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A portfolio website demonstrates all web technologies working together. HTML creates structure with semantic elements. CSS styles with selectors box model and flexbox. JavaScript adds interactivity with DOM manipulation. Responsive design uses media queries and relative units.	Portfolio combines HTML structure CSS style and JavaScript behavior. Responsive design ensures mobile compatibility.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Build your personal portfolio with at least four sections. Include home with photo and introduction about section with education skills section with progress bars and a contact form. Style with external CSS. Add one JavaScript feature.	My portfolio has home about skills and contact sections. CSS flexbox creates layout. JavaScript toggles dark mode.	*Peer tick against Success Criteria*
Evaluate *5 min*	Your project will be evaluated on proper HTML structure correct CSS styling JavaScript functionality responsive design and documentation.	My project uses semantic HTML organized CSS and functional JavaScript. It works on all screen sizes.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Build a simplified portfolio with HTML and CSS only using the... **On** Class 9 Application: Build the complete portfolio with HTML CSS and basic JavaScript... **Adv** Class 10 Mastery: Build an advanced portfolio with multiple JavaScript features and... **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a personal portfolio page about yourself with hobbies favorite subjects and goals. Spiral: also recall one fact from last week.

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Lesson 5.01 · debug · 40 minutes

What is an Algorithm?

I can Analyze an algorithm as a step-by-step procedure to solve a problem and justify your choice · Analyze and compare algorithms in everyday activities like cooking and directions · Write simple algorithms for real-world tasks using clear sequential steps ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students when you make tea you follow specific steps. That recipe is an algorithm. An algorithm is a step-by-step procedure to solve a problem. Today we learn to think algorithmically.	So an algorithm is like a recipe. If the steps are clear anyone can follow them and get the same result.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Write down every step for making chai. Include every detail. Compare with your partner. What happens if steps are missing or in wrong order.	My steps were take water add sugar add tea boil strain and serve. My partner missed boiling. Order matters.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	An algorithm is a finite set of unambiguous steps that solve a specific problem. Key properties are finite unambiguous and effective. Algorithms exist everywhere from recipes to traffic rules. A good algorithm produces correct output for all valid inputs.	Algorithms are finite clear and doable. They exist everywhere. Good algorithms give correct output and run efficiently.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Write algorithms for: tie shoelaces pack school bag and find a book in the library. Each must have at least 5 steps. Test by having a partner follow exactly.	My shoelace algorithm has 7 steps. My bag packing checks the timetable first. My partner followed each successfully.	*Peer tick against Success Criteria*
Evaluate *5 min*	Write a complete algorithm for a given task. Check if steps are finite clear and produce correct result.	Good algorithms are specific enough that anyone can follow them.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Write algorithms for simple tasks like brushing teeth. **On** Class 9 Application: Write algorithms for complex tasks with decision points. **Adv** Class 10 Mastery: Write algorithms with conditions and loops and analyze efficiency. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write an algorithm for your evening study routine from sitting down to going to bed. Spiral: also recall one fact from last week.

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LESSON 5.02 · DEBUG · 40 MINUTES

Flowcharts

I CAN Draw flowcharts using standard symbols for start end process and decision · Convert a written algorithm into a visual flowchart representation · Read and interpret flowcharts to understand program logic ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Algorithms can be drawn as diagrams called flowcharts. Think of a map with directions. The flowchart shows the path your program takes from start to finish using different shapes.	Flowcharts are like maps for programs. The shapes tell me what type of action happens at each step.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Standard symbols: oval for start end, rectangle for process, diamond for decision with yes/no branches, arrows for flow. Draw a flowchart for making tea.	I used oval for start and end, rectangles for boiling and straining, diamond for is it ready. Arrows connect everything.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Flowcharts use standard symbols. Oval represents start and end. Rectangle represents process. Diamond represents decision with two exits. Parallelogram represents input or output. Arrows show flow direction. Flowcharts make algorithms visual and help identify errors before coding.	Oval is start end. Rectangle is process. Diamond is decision. Parallelogram is input output. Flowcharts visualize algorithms.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Convert your algorithm for finding the largest of three numbers into a flowchart. Use proper symbols. Trace through with test values 5 8 3 to verify.	My flowchart starts with input three numbers. First diamond compares A and B. Second compares larger with C. Tracing confirms it works.	*Peer tick against Success Criteria*
Evaluate *5 min*	Draw a flowchart for a given algorithm and describe what a given flowchart does.	Converting between algorithm and flowchart requires understanding the logic.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Draw simple flowcharts with linear steps and one decision. **On** Class 9 Application: Draw flowcharts with multiple decisions and loops. **Adv** Class 10 Mastery: Create complex flowcharts and analyze efficiency. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a flowchart for making your bed from waking up to finishing. Spiral: also recall one fact from last week.

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LESSON 5.03 · DEBUG · 40 MINUTES

Pseudocode

I CAN Write pseudocode to express algorithms in a structured but readable format · Use pseudocode keywords like BEGIN END IF THEN ELSE and WHILE DO · Convert between pseudocode and flowchart representations ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students pseudocode is like writing an algorithm in a format that looks like code but uses plain English. It is halfway between a flowchart and actual programming.	Pseudocode is structured English that describes algorithms. It uses keywords like IF and WHILE but does not follow strict syntax rules.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Let me write pseudocode for finding the largest of three numbers. BEGIN input A B C. IF A greater than B AND A greater than C THEN output A. ELSE IF B greater than C THEN output B. ELSE output C. END.	My pseudocode clearly shows the decision structure. IF-THEN-ELSE handles different cases. BEGIN and END mark the boundaries.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Pseudocode is an informal high-level description of an algorithm. It uses keywords like BEGIN END IF THEN ELSE WHILE DO FOR and RETURN. It does not follow strict syntax so it focuses on logic not syntax. Pseudocode bridges the gap between human language and programming.	Pseudocode uses keywords in plain English. It bridges human language and programming.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write pseudocode for: calculate average of three marks, find if a number is even or odd, and sort three numbers.	My average algorithm computes sum then divides by 3. My even-odd uses MOD operator. My sorting uses nested IF statements.	*Peer tick against Success Criteria*
EVALUATE *5 min*	Convert a given flowchart to pseudocode and verify both represent the same logic.	Pseudocode and flowcharts represent the same algorithms in different formats.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Write pseudocode for linear algorithms using BEGIN END and simple… **On** Class 9 Application: Write pseudocode with IF THEN ELSE and loops. **Adv** Class 10 Mastery: Write complex pseudocode with nested structures and functions. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write pseudocode for your morning routine from alarm to leaving for school. Spiral: also recall one fact from last week.

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LESSON 5.04 · DEBUG · 40 MINUTES

Sequential Algorithms

I CAN Analyze and compare sequential algorithms where steps follow one after another in order · Write sequential algorithms for problems with a definite beginning and end · Trace through sequential algorithms step by step to verify correctness ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Today we focus on sequential algorithms where steps happen in a fixed order. Think of boarding a bus: show ticket find seat sit down. Each step follows the previous one.	Sequential algorithms are the simplest type. Steps follow in order.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Write a sequential algorithm to calculate total marks for 5 subjects. Input each mark then add them all and output total. Trace with sample values.	I input marks 85 90 78 92 88. I added step by step. The total is 433.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Sequential algorithms have steps that execute in order from first to last. There are no decisions or repetitions. They are the foundation for all other algorithm types.	Sequential steps run in fixed order. No decisions or loops. They form the foundation.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write sequential algorithms for: calculate area of rectangle, convert Celsius to Fahrenheit, and calculate distance given speed and time.	My area algorithm multiplies length times width. My temperature uses F equals C times 9/5 plus 32.	*Peer tick against Success Criteria*
EVALUATE *5 min*	Given a problem write a sequential algorithm and trace it with test values.	Sequential algorithms are verified by tracing each step in order.	*Individual exit ticket (2 Q) at door*

RESOURCES

DIFFERENTIATION **App** Class 8 Foundation: Trace pre-written sequential algorithms with sample values. **On** Class 9 Application: Write sequential algorithms for mathematical calculations. **Adv** Class 10 Mastery: Write complex sequential algorithms and analyze their steps. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a sequential algorithm to calculate how many hours you studied today. Spiral: also recall one fact from last week.

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LESSON 5.05 · DEBUG · 40 MINUTES

Selection Algorithms

I CAN Analyze and compare selection algorithms that make decisions using IF THEN ELSE · Write algorithms with single and nested selection structures · Trace selection algorithms to determine which path is taken for given inputs ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Sometimes algorithms need to make decisions. If it rains take umbrella else wear sunglasses. Selection algorithms choose between paths based on conditions.	Selection algorithms make choices. IF a condition is true one path is taken. ELSE another path is taken.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Write an algorithm: IF temperature greater than 35 THEN output hot ELSE IF temperature less than 10 THEN output cold ELSE output pleasant. Trace with values 40, 5, and 20.	For 40 it outputs hot. For 5 it outputs cold. For 20 it outputs pleasant. Different inputs take different paths.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Selection algorithms use conditions to choose between paths. IF condition THEN action ELSE alternative. Nested selection adds more conditions. AND requires both true. OR requires one true.	IF-THEN-ELSE chooses paths. Nested IF handles multiple conditions.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Write selection algorithms for: grade a student mark, determine if a year is a leap year, and classify a number as positive negative or zero.	My grading uses ranges. Leap year checks divisibility by 4 and not by 100 or by 400.	*Peer tick against Success Criteria*
EVALUATE *5 min*	Write a selection algorithm and trace it with different inputs to verify all paths.	Tracing with multiple inputs ensures all paths produce correct results.	*Individual exit ticket (2 Q) at door*

RESOURCES Paper and pen · Textbook Chapter 5 Page 5 · our school pseudocode reference card ·

DIFFERENTIATION **App** Class 8 Foundation: Write simple IF-THEN-ELSE algorithms with one condition. **On** Class 9 Application: Write algorithms with nested selections and multiple conditions. **Adv** Class 10 Mastery: Write complex selection algorithms and analyze all possible paths. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a selection algorithm that decides what to wear based on weather temperature. Spiral: also recall one fact from last week.

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Lesson 5.06 · debug · 40 minutes

Iteration Algorithms

I can Analyze and compare iteration as repetition of steps using loops · Write algorithms with FOR loops for known repetitions and WHILE loops for unknown · Trace iteration algorithms to count loop executions and track variable changes ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Sometimes you need to repeat a step multiple times. Do 10 jumping jacks. Sing the prayer three times. Iteration is repetition using loops.	Iteration means repetition. FOR loops repeat a known number of times. WHILE loops repeat until a condition changes.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Write: FOR i from 1 to 5 DO output i times 2. Trace: outputs 2 4 6 8 10. Now WHILE: set count=1 WHILE count less than or equal to 5 DO output count count equals count plus 1.	FOR loop runs exactly 5 times. WHILE loop also runs 5 times but stops when count exceeds 5.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	FOR loops repeat a known number of times. WHILE loops repeat until a condition becomes false. The loop variable must change toward the exit condition to avoid infinite loops. Counter-controlled FOR and condition-controlled WHILE are the two main types.	FOR is for known counts. WHILE is for unknown counts. The variable must change to avoid infinite loops.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Write iteration algorithms for: print multiplication table of a number, sum numbers from 1 to N, and find the largest number in a list of 10 numbers.	My multiplication table uses FOR loop from 1 to 10. My sum uses WHILE adding each number. My largest compares each element.	*Peer tick against Success Criteria*
Evaluate *5 min*	Write an iteration algorithm and trace it counting each loop execution.	Tracing loops requires tracking the counter and output at each iteration.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Write simple FOR loops with known repetition counts. **On** Class 9 Application: Write both FOR and WHILE loops for different scenarios. **Adv** Class 10 Mastery: Write nested loops and analyze loop efficiency. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write an algorithm that counts from 1 to 20 and prints only even numbers. Spiral: also recall one fact from last week.

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LESSON 5.07 · DEBUG · 40 MINUTES

Big-O Notation

I CAN Understand Big-O notation as a way to describe algorithm efficiency · Analyze and compare common time complexities: O(1) O(n) O(n squared) and O(log n) · Compare algorithms based on their growth rate and efficiency for large inputs ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students not all algorithms are equally fast. Some handle 100 items quickly but struggle with 10000. Big-O notation measures how an algorithm grows as input gets bigger. Think of it like measuring how long a journey takes as the distance increases.	Big-O tells us how fast an algorithm runs. O(1) is constant time. O(n) grows with input. O(n squared) grows much faster.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Compare two search methods on a list of 100 items. Linear search checks up to 100 items. Binary search checks about 7 items. For 1000 items linear checks 1000 but binary checks about 10. The difference grows as input grows.	Linear search is O(n). Binary search is O(log n). For large inputs binary is much faster.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Big-O describes the worst-case growth rate. O(1) constant means same time regardless of input. O(n) linear means time grows with input. O(n squared) quadratic means time grows with input squared. O(log n) logarithmic means time grows very slowly.	O(1) is fastest. O(log n) is fast. O(n) is moderate. O(n squared) is slow for large inputs.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Analyze the time complexity of: checking every element in a list, using binary search on a sorted list, and comparing every pair in a list.	Checking every element is O(n). Binary search is O(log n). Comparing every pair is O(n squared).	*Peer tick against Success Criteria*
Evaluate *5 min*	Given an algorithm determine its Big-O notation and explain why.	Big-O depends on how many operations the algorithm performs relative to input size.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Understand that some algorithms are faster than others for large... **On** Class 9 Application: Identify O(1) O(n) and O(n squared) for simple algorithms. **Adv** Class 10 Mastery: Analyze time complexity of algorithms and compare efficiency. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Compare the efficiency of finding a word in a dictionary by checking every page versus opening to the middle. Spiral: also recall one fact from last week.

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Lesson 5.08 · debug · 40 minutes

Divide and Conquer

I can Understand divide and conquer as breaking a problem into smaller subproblems · Analyze and compare how merge sort and binary search use divide and conquer strategy · Compare divide and conquer approaches with brute force approaches ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students divide and conquer means breaking a big problem into smaller easier pieces solve each piece and combine the results. Think of dividing class work among groups. Each group handles a part and you combine the answers.	Divide and conquer breaks hard problems into easy pieces. Each piece is solved independently. Results are combined.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Binary search divides the search range in half each time. Merge sort divides the list in half sorts each half then merges. Both are faster than checking every element. Draw the steps for merge sort on 8 numbers.	Merge sort divides 8 into two 4s, each into two 2s, each into two 1s. Then merges back sorted. It takes about 24 operations instead of 64 for bubble sort.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Divide and conquer algorithms break a problem into subproblems solve each recursively and combine results. They are often more efficient than brute force. Binary search merge sort and quicksort all use this strategy.	Divide into subproblems, solve each, combine results. More efficient than brute force.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Apply divide and conquer to find the largest element in a list: split the list in half find the largest in each half then compare the two largest. Trace with 8 numbers.	Split 8 into two 4s. Find largest in each. Compare the two winners. Much fewer comparisons than checking all 8.	*Peer tick against Success Criteria*
Evaluate *5 min*	Analyze and compare whether a given algorithm uses divide and conquer and explain the subproblem structure.	Divide and conquer algorithms have clear divide solve and combine steps.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Understand the concept of dividing a problem into smaller parts. **On** Class 9 Application: Trace merge sort and binary search step by step. **Adv** Class 10 Mastery: Analyze divide and conquer efficiency and compare with other approaches. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Use divide and conquer to find the median of a list of 6 numbers. Spiral: also recall one fact from last week.

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LESSON 5.09 · DEBUG · 40 MINUTES

Greedy Algorithms

I CAN Understand greedy algorithms that make the best local choice at each step · Apply greedy strategy to problems like making change and activity selection · Analyze and compare when greedy algorithms work well and when they fail ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students greedy algorithms make the best choice right now without worrying about the future. Like getting the best deal at each shop while shopping. Sometimes this gives the best overall result but not always.	Greedy algorithms pick the best local option. Sometimes this gives the global best. Sometimes it does not.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Make change for 37 rupees using the fewest coins. Greedy approach: use the largest coin first. 37 = 25 + 10 + 1 + 1 = 4 coins. Is this always the best? Try making change for a currency where greedy fails.	For Indian currency greedy works well. But for some coin systems it fails. Greedy is simple but not always optimal.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Greedy algorithms make the locally optimal choice at each step hoping for a globally optimal result. They work well for some problems like activity selection and Huffman coding. They fail for problems like the traveling salesman.	Greedy picks the best local choice. Works for some problems. Fails for others.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Apply greedy to: schedule maximum activities from a list with start and end times, and make change for 63 rupees with minimum coins.	For activities I sort by end time and pick non-overlapping ones. For change 63 = 50 + 10 + 1 + 1 + 1 = 5 coins.	*Peer tick against Success Criteria*
Evaluate *5 min*	Determine if a greedy approach works for a given problem and justify your answer.	Greedy works when local optimal choices lead to global optimal solution.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Understand the concept of making the best local choice. **On** Class 9 Application: Apply greedy to simple problems like making change. **Adv** Class 10 Mastery: Analyze when greedy works and when it fails with proof. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Use a greedy approach to pack your bag placing the most important items first. Spiral: also recall one fact from last week.

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Lesson 5.10 · debug · 40 minutes

Algorithm Efficiency

I can Compare the efficiency of different algorithms for the same problem · Understand that efficiency affects how programs handle large datasets · Choose the most efficient algorithm based on problem requirements ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students efficiency matters when working with large data. An efficient algorithm on a slow computer beats an inefficient algorithm on a fast computer. Today we compare algorithm efficiencies.	Efficiency determines how well an algorithm scales. Choosing the right algorithm is more important than faster hardware.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Compare bubble sort O(n squared) with binary search O(log n) on a list of 10000 items. Bubble sort does about 100 million operations. Binary search does about 14. The difference is enormous.	Bubble sort: 10000 squared equals 100 million. Binary search: log of 10000 is about 14. Binary search is millions of times faster.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Efficiency is measured by time complexity and space complexity. Time complexity measures how operations grow. Space complexity measures memory usage. The best algorithm depends on data size structure and available resources.	Time and space complexity measure efficiency. Best algorithm depends on the specific problem.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Compare three algorithms for finding duplicates in a list: check every pair O(n squared) sort first O(n log n) and use a set O(n). Discuss when each is appropriate.	Check every pair is simple but slow. Sort is faster. Set is fastest but uses more memory.	*Peer tick against Success Criteria*
Evaluate *5 min*	For a given problem compare different algorithm approaches and recommend the most efficient one with justification.	The best algorithm depends on data size, memory availability, and code complexity.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Understand that some algorithms are faster than others. **On** Class 9 Application: Compare algorithm efficiencies for small and large inputs. **Adv** Class 10 Mastery: Analyze time and space complexity and choose optimal algorithms. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Compare two ways to find your name in a phone directory: page by page versus binary search. Spiral: also recall one fact from last week.

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Lesson 5.11 · debug · 40 minutes

Problem Decomposition

I can Break complex problems into smaller manageable subproblems · Analyze and compare how smaller solutions combine to solve the original problem · Apply decomposition to real-world our school scenarios ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students complex problems are overwhelming when viewed as a whole. Decomposition breaks them into smaller pieces. Like eating an elephant one bite at a time. Each subproblem is easier to solve.	Decomposition makes hard problems manageable. Each small piece is solved separately then combined.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Design a student report card system. Decompose into: input student data, calculate averages, determine grades, generate report. Each subproblem is simpler than the whole.	My system has four subproblems. Each can be solved independently. Together they produce the complete report card.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Decomposition is breaking a complex problem into simpler subproblems. Each subproblem can be solved independently. Subproblems are then combined to solve the original problem. This is a fundamental problem-solving strategy in computing.	Break complex into simple. Solve each independently. Combine solutions.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Decompose these our school problems: manage library system, plan school event, and create online quiz. Identify at least 3 subproblems for each.	Library system needs catalog, borrowing, and returns. Event needs venue, schedule, and invitations.	*Peer tick against Success Criteria*
Evaluate *5 min*	Given a complex problem decompose it into subproblems and describe how solving each contributes to the whole.	Good decomposition creates independent subproblems that cover the entire problem.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Decompose simple problems into 2-3 subproblems. **On** Class 9 Application: Decompose complex problems and solve each subproblem. **Adv** Class 10 Mastery: Apply decomposition to large projects with dependencies. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Decompose the problem of organizing a class picnic into subproblems. Spiral: also recall one fact from last week.

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Lesson 5.12 · debug · 40 minutes

Algorithm Design Project

I can Apply algorithmic thinking to design a complete solution for a real problem · Use flowcharts pseudocode and efficiency analysis in the design process · Document the design process and justify algorithmic choices ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students this is our final algorithm project. You will use everything learned: algorithms flowcharts pseudocode selection iteration decomposition and efficiency analysis to solve a real problem.	I need to plan my solution using algorithms choose efficient approaches and document everything clearly.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Design a residential canteen queue management system. Identify the problems to solve: take orders, manage queue, calculate bills, track inventory. Design algorithms for each. Create flowcharts. Analyze efficiency.	My system has four algorithms. The queue uses FIFO. Bill calculation uses selection for discounts. Inventory uses iteration to check stock levels.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A complete algorithm design project follows these steps: understand the problem, decompose into subproblems, design algorithms for each, represent with flowcharts and pseudocode, analyze efficiency, and document the process.	Project steps: understand decompose design represent analyze document.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Design a complete solution for: online homework submission system or digital attendance tracker. Include all design steps from decomposition to efficiency analysis.	My attendance system decomposes into: mark attendance view records generate report. Each has efficient algorithms.	*Peer tick against Success Criteria*
Evaluate *5 min*	Present your design explaining each algorithm choice and efficiency analysis.	Good design documentation explains why choices were made not just what was chosen.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Design a simplified system with 2-3 algorithms. **On** Class 9 Application: Design a complete system with flowcharts and pseudocode. **Adv** Class 10 Mastery: Design with efficiency analysis and comprehensive documentation. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Design an algorithm for a simple calculator that handles + - * / operations. Spiral: also recall one fact from last week.

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LESSON 6.01 · DEBUG · 40 MINUTES

What is AI?

I CAN Analyze artificial intelligence as machines that can perform tasks requiring human intelligence and justify... · Analyze and compare examples of AI in daily life including voice assistants and recommendation systems · Distinguish between narrow AI and general AI capabilities ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students you have seen robots in movies that think and act like humans. Today we learn about artificial intelligence or AI which is making machines smart. AI is already around you in your phone assistant and video recommendations.	AI is making computers smart. My phone recognizes my face and my voice assistant answers questions. AI is already part of my daily life.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	List five examples of AI you use: voice assistants like Alexa face unlock on phones recommendation videos on YouTube spell check in documents and auto-complete in messages. Discuss how each uses intelligence.	My examples are Siri face recognition YouTube recommendations Google spell check and WhatsApp auto-complete. Each mimics a human ability like seeing hearing or predicting.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Artificial intelligence is machines performing tasks that normally require human intelligence. This includes learning from data recognizing patterns making decisions and understanding language. AI works by training on large datasets to find patterns. Narrow AI handles specific tasks like voice recognition. General AI would handle any intellectual task like a human.	AI mimics human intelligence. It learns from data and finds patterns. Narrow AI handles specific tasks. General AI does not yet exist.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Research one AI application used in India. Explain how it works and who benefits from it. Consider agriculture healthcare education or banking.	I researched AI in Indian agriculture. It analyzes soil and weather data to advise farmers. This helps farmers grow better crops with less waste.	*Peer tick against Success Criteria*
EVALUATE *5 min*	For each AI example identify the human intelligence it mimics and discuss benefits and risks.	AI mimics human abilities like seeing hearing and deciding. Benefits include convenience. Risks include privacy concerns.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Focus on recognizing AI in everyday life and understanding what AI... **On** Class 9 Application: Analyze how AI works in specific applications and discuss benefits... **Adv** Class 10 Mastery: Compare narrow and general AI and evaluate the ethical implications of... **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List five AI tools you use this week and describe what each does for you. Spiral: also recall one fact from last week.

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LESSON 6.02 · DEBUG · 40 MINUTES

Machine Learning Basics

I CAN Explain machine learning as AI that learns from data without being explicitly programmed · Analyze and compare three types of machine learning: supervised unservised and reinforcement · Understand how training data is used to build machine learning models ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students machine learning is a type of AI where computers learn from examples instead of being told every rule. Like how you learn to recognize animals by seeing many animals. Today we learn how computers learn.	Machine learning means computers learn from data. Instead of writing rules for every situation I give the computer examples and it learns the patterns.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Show a simple example: sorting fruits by weight and color. Give the computer 100 examples of apples and oranges with their weight and color. The computer finds the pattern: apples tend to be smaller and redder oranges are larger and orange. Now it can sort new fruits.	The computer learned the pattern from examples. Weight and color help distinguish apples from oranges. More examples make the computer more accurate.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Machine learning is AI that improves through experience. Supervised learning uses labeled training data with correct answers. Unsupervised learning finds hidden patterns in unlabeled data. Reinforcement learning learns through trial and error with rewards.	Supervised uses labeled data. Unsupervised finds hidden patterns. Reinforcement learns through rewards.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Describe a machine learning scenario for our school: predicting which students might need extra help based on past performance. What data would you use and what type of learning is this.	This is supervised learning. I would use past marks attendance and homework data. The model learns which patterns predict low performance.	*Peer tick against Success Criteria*
Evaluate *5 min*	For a given scenario identify whether it uses supervised unsupervised or reinforcement learning and justify your choice.	The type depends on whether data has labels and whether there is a reward system.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Understand that computers can learn from examples and data. **On** Class 9 Application: Identify the three types of machine learning and give examples of... **Adv** Class 10 Mastery: Analyze how machine learning models are trained and discuss bias in... **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Give three examples of machine learning from your daily life and classify each type. Spiral: also recall one fact from last week.

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LESSON 6.03 · DEBUG · 40 MINUTES

Neural Networks

I CAN Understand neural networks as computing systems inspired by the human brain · Explain how neural networks process information through layers of connected nodes · Analyze and compare applications of neural networks in image recognition and language processing ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students your brain has billions of neurons connected to each other. A neural network is a computer system inspired by this. Each artificial neuron receives inputs processes them and sends outputs to other neurons.	Neural networks are like digital brains. They have layers of connected nodes. Each node processes information and passes it forward.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Draw a simple neural network with 3 input nodes 2 hidden nodes and 1 output node. Input values flow through connections with weights. Each hidden node calculates a weighted sum. The output node produces the final result.	My network has inputs feeding into hidden nodes which feed into output. Weights determine how much each input matters. The output changes as weights change.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Neural networks are layers of interconnected nodes inspired by biological neurons. Input layer receives data. Hidden layers process it. Output layer produces results. Networks learn by adjusting connection weights during training.	Input-hidden-output layers. Weights adjust during training. Networks learn patterns from data.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Research how neural networks are used in handwriting recognition. Explain the layers involved and how the network learns to identify different letters.	Handwriting recognition uses input nodes for pixel values. Hidden layers detect features like curves and lines. Output nodes identify the letter. Training adjusts weights until recognition is accurate.	*Peer tick against Success Criteria*
EVALUATE *5 min*	Explain how a neural network processes information from input to output for a given scenario.	Data flows through layers. Weights determine importance. Output represents the prediction.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Understand the concept of layers and how information flows through a... **On** Class 9 Application: Draw simple neural networks and trace data flow through the layers. **Adv** Class 10 Mastery: Explain how neural networks learn through weight adjustment and... **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a neural network that decides if a student will pass or fail based on marks and attendance inputs. Spiral: also recall one fact from last week.

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LESSON 6.04 · DEBUG · 40 MINUTES

IoT Introduction

I CAN Analyze Internet of Things as everyday objects connected to the internet to collect and share data and… · Analyze and compare IoT devices in smart homes schools and cities · Understand how IoT sensors collect data and send it for processing ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students imagine your fan automatically turns on when the room gets hot. Your water bottle tells you how much water you drank. This is IoT the Internet of Things where everyday objects connect to the internet.	IoT means connecting everyday objects to the internet. Smart fans water bottles and watches can all collect data and communicate.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	List IoT devices in a smart home: smart bulbs that adjust brightness smart fans that detect temperature smart meters that track electricity and security cameras that send alerts. Each device has a sensor and internet connection.	Smart bulbs have light sensors. Smart fans have temperature sensors. Smart meters track usage. Cameras have motion sensors. All send data to the internet.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	IoT connects physical objects to the internet. Devices have sensors to collect data processors to analyze it and network connections to share it. Data is often processed in the cloud. IoT enables smart homes cities and industries.	Sensors collect data. Processors analyze it. Network shares it. Cloud stores and processes it.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Design a smart school system using IoT. Identify what devices would be connected what sensors they would have and what data they would collect.	My smart school has attendance sensors at the door temperature sensors in classrooms and energy sensors on lights. All data goes to a central dashboard.	*Peer tick against Success Criteria*
Evaluate *5 min*	For a given scenario identify the IoT devices sensors data collected and benefits.	IoT requires sensors connectivity and data processing to provide benefits.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Identify IoT devices and understand they collect and share data. **On** Class 9 Application: Design simple IoT systems with sensors data collection and benefits. **Adv** Class 10 Mastery: Analyze IoT security challenges and design comprehensive IoT solutions. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List five IoT devices that could improve life at our school and explain what data each would collect. Spiral: also recall one fact from last week.

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LESSON 6.05 · DEBUG · 40 MINUTES

Smart Devices

I CAN Understand how smart devices use sensors processors and connectivity to automate tasks · Compare smart devices with traditional devices and identify the advantages · Analyze the impact of smart devices on daily life at our school and in homes ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE *5 min*	Students smart devices are like regular devices with a brain. A regular fan just spins. A smart fan senses temperature and adjusts speed. Today we explore how smart devices work and their impact.	Smart devices add intelligence to regular devices. They sense their environment make decisions and act automatically.	*Vocabulary pre-taught; prior ideas on board*
EXPLORE *10 min*	Compare a regular alarm clock with a smart alarm. Regular just rings at set time. Smart analyzes sleep patterns wakes you during light sleep and adjusts based on your schedule. The smart version adapts to your needs.	Regular alarm is simple and fixed. Smart alarm adapts uses data and provides personalized experience.	*Circulate; verify role rotation every 7 min*
EXPLAIN *10 min*	Smart devices have four components: sensors to collect data processors to analyze it connectivity to share data and actuators to take action. This creates a cycle: sense decide act communicate. Smart homes schools and cities use networks of these devices.	Sensors process data. Processors decide. Actuators act. Connectivity communicates. Sense-decide-act-communicate cycle.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
ELABORATE *10 min*	Analyze the impact of smart devices at our school: smart attendance smart classroom and smart library. For each identify the technology used benefits and potential concerns.	Smart attendance uses facial recognition benefits include accuracy concerns include privacy. Smart classroom uses interactive boards benefits include engagement.	*Peer tick against Success Criteria*
EVALUATE *5 min*	Evaluate the benefits and drawbacks of adopting smart devices in a school environment.	Benefits include efficiency and personalization. Drawbacks include cost privacy and dependency.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Understand what makes a device smart and identify examples. **On** Class 9 Application: Analyze smart device components and their functions. **Adv** Class 10 Mastery: Evaluate the societal impact of smart device adoption. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a report on how three smart devices could improve life at our school. Spiral: also recall one fact from last week.

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LESSON 6.06 · DEBUG · 40 MINUTES

Cloud Computing Advanced

I CAN Understand cloud computing as storing and accessing data and programs over the internet · Compare IaaS PaaS and SaaS cloud service models with real-world examples · Analyze the benefits of cloud computing for schools including cost and scalability ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students instead of storing everything on your computer you can store it on the internet called the cloud. Google Drive YouTube and Gmail all use cloud computing. Today we learn how the cloud works.	Cloud computing means using internet servers instead of your own device. I can access my files from any device anywhere.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Compare cloud services: Gmail is SaaS you use it directly. Google App Engine is PaaS you build apps on it. AWS is IaaS you rent virtual computers. Each gives you different levels of control.	SaaS gives ready-to-use apps. PaaS gives development platforms. IaaS gives raw computing resources. Each level adds more control and responsibility.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Cloud computing delivers computing services over the internet. IaaS provides virtual machines and storage. PaaS provides development platforms. SaaS provides ready-to-use applications. Benefits include cost savings scalability and accessibility.	Cloud is computing over internet. IaaS PaaS SaaS are service levels. Benefits are cost scalability accessibility.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Analyze how our school could use cloud computing: Google Classroom for LMS Google Drive for file storage and Zoom for online classes. For each identify the service model and benefits.	Google Classroom is SaaS for learning. Google Drive is SaaS for storage. Zoom is SaaS for communication. Benefits include accessibility and collaboration.	*Peer tick against Success Criteria*
Evaluate *5 min*	Compare cloud computing with local computing and recommend when each is appropriate.	Cloud is better for collaboration and accessibility. Local is better for speed and privacy.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Understand that cloud computing stores data on internet servers. **On** Class 9 Application: Compare IaaS PaaS and SaaS with real-world examples. **Adv** Class 10 Mastery: Analyze cloud computing costs benefits and security considerations. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List five cloud services you use and identify whether each is IaaS PaaS or SaaS. Spiral: also recall one fact from last week.

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Lesson 6.07 · debug · 40 minutes

Blockchain Basics

I can Understand blockchain as a distributed digital ledger that records transactions securely · Explain how blockchain ensures data cannot be altered once recorded · Analyze and compare applications of blockchain beyond cryptocurrency including supply chain and voting ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students blockchain is like a digital ledger that everyone can see but no one can change. Originally designed for Bitcoin it now has many uses. Think of it as a chain of blocks where each block contains records.	Blockchain is a permanent digital record. Once something is recorded it cannot be changed. This makes it trustworthy.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Demonstrate blockchain: write a transaction on a block connect it to the previous block with a hash. Try to change one block and observe that all following blocks become invalid. This shows immutability.	Each block has a unique hash. Changing one block changes its hash which breaks the chain. This makes blockchain tamper-proof.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Blockchain is a distributed ledger across many computers. Each block contains transactions and a hash linking to the previous block. Once recorded data cannot be altered. This creates trust without a central authority.	Distributed ledger across computers. Blocks contain transactions and hashes. Immutable once recorded. Trust without central authority.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Research three non-cryptocurrency applications of blockchain: supply chain tracking medical records and digital identity. Explain how blockchain benefits each application.	Supply chain tracks products from farm to table. Medical records ensure data integrity. Digital identity prevents fraud.	*Peer tick against Success Criteria*
Evaluate *5 min*	Explain how blockchain achieves trust and security without a central authority.	Distributed storage consensus mechanisms and cryptographic hashing create trust.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Understand blockchain as a permanent digital record that cannot be... **On** Class 9 Application: Explain how blocks are linked and why changing one breaks the chain. **Adv** Class 10 Mastery: Analyze blockchain applications and evaluate their benefits and... **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a simple blockchain with 5 blocks and show what happens when someone tries to change block 3. Spiral: also recall one fact from last week.

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LESSON 6.08 · DEBUG · 40 MINUTES

Quantum Computing Intro

I CAN Understand quantum computing as a new type of computing using quantum mechanical phenomena · Explain the concepts of qubits superposition and entanglement in simple terms · Compare quantum computing potential with classical computing limitations ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students classical computers use bits that are 0 or 1. Quantum computers use qubits that can be 0 and 1 at the same time called superposition. This makes them potentially much more powerful for certain problems.	Quantum computers use qubits instead of bits. Qubits can be 0 and 1 simultaneously. This allows exploring many solutions at once.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Compare classical and quantum approaches to finding a word in a dictionary. Classical checks one page at a time. Quantum could theoretically check all pages simultaneously using superposition.	Classical checks pages one by one. Quantum explores many pages at once. Quantum is exponentially faster for some problems.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Quantum computing uses qubits which exploit superposition being in multiple states at once and entanglement where qubits are connected. This enables solving certain problems exponentially faster than classical computers.	Qubits use superposition and entanglement. Quantum solves some problems exponentially faster than classical.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Research one real-world problem that quantum computing could solve: drug discovery climate modeling or cryptography. Explain why classical computers struggle with this problem.	Drug discovery requires simulating molecular interactions. Classical computers cannot handle the complexity. Quantum computers could simulate molecules accurately.	*Peer tick against Success Criteria*
Evaluate *5 min*	Explain the potential advantages and current limitations of quantum computing.	Quantum is faster for specific problems but currently expensive fragile and limited in availability.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Understand that quantum computers work differently from classical... **On** Class 9 Application: Explain superposition and entanglement concepts in simple terms. **Adv** Class 10 Mastery: Compare quantum and classical computing capabilities and limitations. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Explain in your own words why quantum computers could be faster than classical computers for certain tasks. Spiral: also recall one fact from last week.

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LESSON 6.09 · DEBUG · 40 MINUTES

Ethics in Technology

I CAN Analyze and compare ethical issues in technology including privacy bias and fairness · Understand the importance of responsible technology development and use · Analyze real-world scenarios involving technology ethics and propose solutions ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students technology is powerful but must be used responsibly. Today we discuss ethics in technology. Questions like who owns your data should AI make decisions about people and is technology fair to everyone.	Technology ethics means using technology responsibly. I need to think about fairness privacy and the impact of technology on people.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Discuss these scenarios: an AI hiring system that discriminates against women a social media app that sells user data and a facial recognition system that works better for some races. Identify the ethical issues.	The hiring AI shows bias. The social media app violates privacy. The facial recognition is unfair. All involve ethical issues that affect real people.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Technology ethics covers privacy fair access algorithmic bias intellectual property and environmental impact. Developers have responsibility to create fair transparent and safe technologies. Users have responsibility to use technology ethically.	Privacy fair access bias transparency responsibility. Both developers and users have ethical obligations.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Analyze an ethical dilemma: Should schools use facial recognition for attendance? Consider privacy accuracy fairness and alternatives. Present both sides of the argument.	Benefits include accuracy and automation. Risks include privacy violation and data security. Alternatives include ID cards.	*Peer tick against Success Criteria*
Evaluate *5 min*	Propose ethical guidelines for technology use at our school considering privacy fairness and responsibility.	Good guidelines address data collection consent fairness and accountability.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Identify basic ethical issues in technology use. **On** Class 9 Application: Analyze ethical scenarios and discuss both sides. **Adv** Class 10 Mastery: Evaluate technology ethics and propose guidelines for responsible use. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write three ethical guidelines for using technology at our school. Spiral: also recall one fact from last week.

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Lesson 6.10 · debug · 40 minutes

Digital Divide

I can Analyze digital divide as the gap between those with and without access to technology and justify your choice · Analyze and compare causes and consequences of the digital divide in India and globally · Propose solutions to bridge the digital divide and ensure equitable access ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students not everyone has equal access to technology. Some have smartphones and internet. Others have neither. This gap is called the digital divide. Today we explore why it exists and how to bridge it.	The digital divide means unequal access to technology. Some people have much more technology than others. This creates inequality.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Research the digital divide in Karnataka: urban schools have computer labs while rural schools may not. Urban students have internet at home while rural students may not. List the causes and effects of this gap.	Causes include poverty infrastructure and education. Effects include unequal learning opportunities and job prospects.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	The digital divide has three aspects: access to devices availability of internet and digital literacy. It exists between urban and rural areas wealthy and poor nations and different age groups. Bridging it requires infrastructure education and affordable technology.	Three aspects: devices internet literacy. Exists between urban-rural rich-poor. Requires infrastructure education affordability.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Propose solutions for bridging the digital divide at our school: how can the school ensure all students have equal access to technology for learning.	Solutions include shared computer labs mobile device lending libraries and digital literacy programs.	*Peer tick against Success Criteria*
Evaluate *5 min*	Evaluate the impact of the digital divide on education and propose policy recommendations.	Digital divide creates educational inequality. Policies should address access literacy and affordability.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Understand what the digital divide is and identify examples. **On** Class 9 Application: Analyze causes and effects of the digital divide. **Adv** Class 10 Mastery: Evaluate solutions and propose policies to bridge the divide. **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a proposal for bridging the digital divide in your local community. Spiral: also recall one fact from last week.

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LESSON 6.11 · DEBUG · 40 MINUTES

Future Careers

I CAN Analyze and compare careers emerging from technology including AI specialist data scientist and... · Understand how technology is changing traditional careers and creating new ones · Explore the skills needed for technology careers and plan learning paths ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students technology is creating new careers that did not exist before. AI specialist data scientist cybersecurity expert and app developer are jobs today that were not around 20 years ago. Let us explore future careers.	Technology creates new careers. AI specialist data scientist and cybersecurity are growing fields. I need to prepare for these opportunities.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Research five technology careers: their role required skills and education path. Compare with traditional careers like doctor engineer and teacher. How do technology careers differ.	Technology careers require coding data analysis and continuous learning. They often offer remote work and high demand.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	Technology is transforming all careers. Traditional jobs now require digital skills. New careers emerge from AI data science cybersecurity and renewable technology. Key skills include coding data analysis problem-solving and adaptability.	Technology transforms all careers. New roles emerge. Key skills are coding data analysis problem-solving adaptability.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Create a career exploration plan for yourself: identify three technology careers that interest you the skills required and the steps to prepare for them during and after school.	My plan includes learning Python for AI studying data analysis for data science and understanding networks for cybersecurity.	*Peer tick against Success Criteria*
Evaluate *5 min*	Compare a traditional career path with a technology career path and identify the transferable skills.	Both require communication problem-solving and continuous learning. Technology careers add coding and data skills.	*Individual exit ticket (2 Q) at door*

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DIFFERENTIATION **App** Class 8 Foundation: Identify technology careers and the basic skills they require. **On** Class 9 Application: Research career paths and plan skill development. **Adv** Class 10 Mastery: Create comprehensive career plans with educational pathways. **SEND** Trace code with finger and use visual block cards with teacher support

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Interview a technology professional and write about their career journey. Spiral: also recall one fact from last week.

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Lesson 6.12 · debug · 40 minutes

Technology Impact Essay

I can Write a structured essay analyzing the impact of technology on society · Support arguments with evidence from the technology topics studied in this chapter · Present balanced perspectives on both benefits and risks of technology ·

Phase	Teacher Activity	Student Activity	Assessment Check
Engage *5 min*	Students this is our final project. You will write an essay analyzing technology impact using everything learned about AI IoT cloud computing ethics and the digital divide. This brings all chapter topics together.	I need to organize my knowledge from all chapter topics into a well-structured essay with clear arguments and evidence.	*Vocabulary pre-taught; prior ideas on board*
Explore *10 min*	Plan your essay: introduction body paragraphs on AI IoT ethics and digital divide and conclusion. Use evidence from lessons to support each point. Consider both benefits and risks.	My essay has five paragraphs. Introduction sets context. Body paragraphs cover AI benefits IoT transformation ethics challenges and digital divide concerns. Conclusion balances both perspectives.	*Circulate; verify role rotation every 7 min*
Explain *10 min*	A technology impact essay has a clear thesis statement evidence-based body paragraphs and a thoughtful conclusion. It presents balanced perspectives acknowledging both benefits and risks. Good essays use specific examples and data.	Clear thesis evidence-based paragraphs balanced perspectives specific examples.	*Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min*
Elaborate *10 min*	Write a 500-word essay on technology impact covering at least four chapter topics. Include a thesis statement evidence for each point and a balanced conclusion.	My essay argues that technology brings great benefits but requires responsible use. I covered AI IoT ethics and digital divide with specific examples.	*Peer tick against Success Criteria*
Evaluate *5 min*	Evaluate peer essays for thesis clarity evidence use and balanced perspective.	Good essays have clear arguments supported by evidence from multiple topics.	*Individual exit ticket (2 Q) at door*

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Differentiation **App** Class 8 Foundation: Write a simple essay with introduction body and conclusion on one… **On** Class 9 Application: Write an essay covering three topics with evidence and balanced… **Adv** Class 10 Mastery: Write a comprehensive essay covering all chapter topics with strong… **SEND** Trace code with finger and use visual block cards with teacher support

Dormitory · paper Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write an essay about how technology has changed education in India. Spiral: also recall one fact from last week.

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